# The bias interval of the elephant random walk

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**Abstract**

We prove the conjecture of Guérin, Laulin, Raschel and Simon that the bimodal initial biases of the superdiffusive elephant random walk form a possibly empty open interval centered at one half. Outside it, including its endpoints, the limit density has a unique local maximum. We also prove analytically that fair-start bimodality has a unique memory threshold. A positive-weight integral for the central curvature reduces uniqueness to a comparison of the weights as memory increases. Finite sign bounds pass to the limit under weak convergence and give exactly two inflection points in the unimodal case and four in the bimodal case. Numerical certificates locate the threshold near $0.66881286$ and establish its quartic top. The two modes then separate on a square-root scale, while the bias interval opens on a three-halves scale. Above the threshold, the bias boundary is positive and continuously differentiable, with a variational formula and a unique optimizer. Thus strongly biased starts are unimodal at every fixed memory, although the boundary tends to zero as memory tends to one.

**Keywords:** elephant random walk; bimodality; shape transition; variation diminution; initial bias.
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## 1 Introduction

*An elephant and an ant are playing hide and seek. The elephant counts while the ant hides among many identical anthills, then opens its eyes and knows at once which one the ant is in. The ant's shoes are outside the door.*

(Anonymous)

Fix the first step of a superdiffusive elephant random walk. Guérin, Laulin, Raschel and Simon prove that its limit density is unimodal [8, Theorem 1.1]. Randomizing the first step mixes this density with its reflection. The mixture can have two peaks, even though each component has one. They conjecture that the biases giving two peaks form

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an open interval centered at one half [8, Remark 2.1(b)]. We prove their conjecture and locate the unique memory threshold where two peaks first appear under a fair start. A related conjecture, eventual log-concavity, is disproved by Trinas and Valle in a recent preprint [12].

Guérin, Laulin, Raschel and Simon prove conditional unimodality by induction in the finite walk, then pass to the limit. For an arbitrary start we must allow two peaks. We enlarge each row to include its boundary history (the endpoint values needed at later lengths). The sign argument then covers the ends too. On the half favored by the first step, the induction rules out a valley. Averaging over short lattice intervals passes both shape bounds to the limit under weak convergence (Lemma A.7). For curvature we first smooth with a Gaussian, then use Descartes' rule for exponential sums. The new step is the boundary-history construction and its dominant-half induction, which let us use this classical rule for the biased walk.

### 1.1 The walk and its parameters

The elephant walk [1] takes steps $X_1, \dots, X_N \in \{+1, -1\}$, with $+1$ meaning right. To take the next step, it chooses uniformly from its earlier steps. It repeats the chosen direction, except with probability $\varepsilon$, when it goes the other way. Write $U_t$ for the step chosen at time $t+1$, and call it the parent of that step. We choose all parents and all reversal coins independently, also independently of $X_1$.

Put $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The position is $S_N = 2A_N - N$. We take a fair first step unless another bias is specified, so $P_N(k) = P_N(N-k)$. The notation $\mathbb{P}_+$ and $\mathbb{E}_+$ conditions on $X_1 = 1$. For even $N$, write $k_c = N/2$. All logarithms are natural.

For fixed memory put $a = 1 - 2\varepsilon \in (1/2, 1)$ and condition on $X_1 = 1$. The almost-sure limit of $S_N/N^a$ has density $h_a$. With $\mathbb{P}(X_1 = 1) = q$, its density is

$$g_{a,q}(u) = q h_a(u) + (1-q) h_a(-u), \qquad g_a = g_{a,1/2}.$$

The copying probability in the original notation is $p = (1+a)/2$. In the finite recurrences we use $r_t(k) = \varepsilon + (1-2\varepsilon)k/t$.

### 1.2 The bias interval and its threshold

**Theorem 1.1** (the bias interval, analytic)**.** *For every $a \in (1/2, 1)$ there is $b_a \in (0, 1/2]$ such that $g_{a,q}$ has exactly two local maxima if and only if $b_a < q < 1 - b_a$. Outside this interval, including its endpoints, it is unimodal with a unique local maximum. For a fair start the density has exactly two inflection points when unimodal and four when bimodal. There is exactly one $a_* \in (2/3, 69/100)$ with $g''_{a_*}(0) = 0$, and the bias interval is empty exactly when $a \le a_*$. The central curvature is strictly negative below $a_*$ and strictly positive above it, with $\partial_a g''_a(0)|_{a=a_*} > 0$. On $(a_*, 1)$, the boundary is continuously differentiable, and $b_a \to 0$ as $a \uparrow 1$.*

Theorems 3.1 and 3.5 prove the shape assertions. Theorem 3.3 proves the unique transition using a positive-weight integral for the curvature. Theorem 3.7 proves positivity and regularity where the interval is nonempty, and Corollary 3.8 gives the limit at one. None of these arguments needs a numerical certificate.

**Theorem 1.2** (the critical shape, computer-assisted)**.** *The threshold satisfies $a_* \approx 0.6688$. At $a_*$ the fair density has a quartic top. Its two modes emerge at distance proportional to $\sqrt{a - a_*}$, and*

$$\tfrac{1}{2} - b_a \sim Q_*(a - a_*)^{3/2}, \qquad Q_* \approx 9.86698769.$$

*The interval strictly widens immediately above $a_*$.*

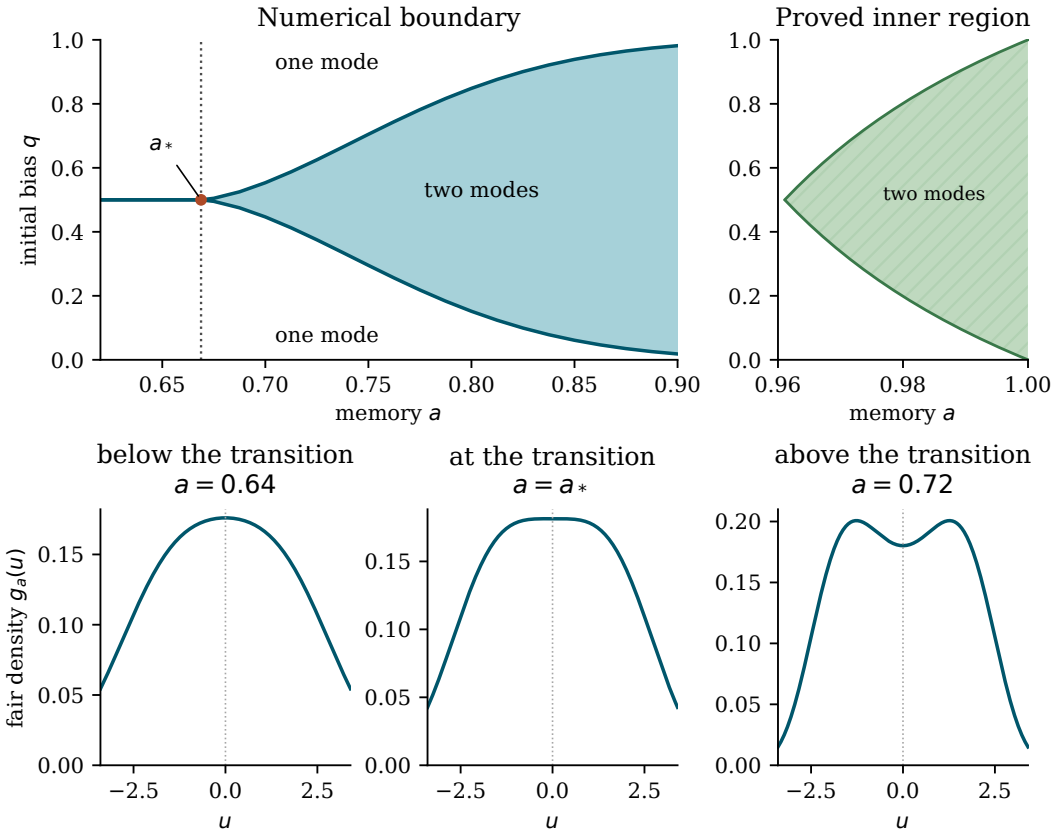



Figure 1: The bias interval and the fair-start transition. The upper left panel shows numerical values of the continuous boundary in Theorem 3.7 (the curves and shading are not certified enclosures). The upper right panel plots the explicit inner region $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$ proved in Corollary 3.8. It does not locate the full boundary. The lower panels show the fair density before, at and after the transition. Its top is quartic at the threshold, whose certified enclosure is given in Appendix C.

The certified enclosure is in Appendix C. The certificates locate the zero and establish the derivative bounds in Table 1. The quartic and three-halves laws then follow by Taylor expansion (Propositions 3.9 and 3.10). Theorem 3.7 gives a continuous boundary, continuously differentiable and positive on $(a_*, 1)$, and a unique optimizer in its variational formula. Continuity at $a_*$ uses the certified critical expansion. We do not prove that the boundary decreases throughout $(a_*, 1)$.

**Theorem 1.3** (the finite shape bounds)**.** *For every $N \geq 1$, every $0 < \varepsilon < 1$ and every initial bias, the count law has at most two modal plateaus, including its endpoints. If $q \geq 1/2$, it dominates its reflection on the right half and has no strict valley there. For a fair start, $N \geq 3$ and $0 < \varepsilon < 1/2$, the nonzero signs of*

$$D_N(k) = P_N(k+1) - 2P_N(k) + P_N(k-1), \qquad 1 \leq k < N,$$

*after equal neighbors are merged, form a subsequence of $(+, -, +, -, +)$.*

The proofs are Proposition A.5, Proposition A.6 and Theorem 2.5. Weak convergence is enough to carry the valley bounds to a continuous limit (Lemma A.7). For the curvature bound, we first smooth by a Gaussian and then apply the sign bound.

### 1.3 How the proof works

**Signs at the ends.** A new sign can enter at an end, where the interior recurrence does not apply. We keep the endpoint values in the enlarged row before they are needed. Each update now either copies an entry or takes a positive sum of neighbors, so the sign bound holds for the whole row. For first differences this gives at most two modal plateaus. For second differences under a fair start it gives at most five sign blocks.

**The half favored by the first step.** Take $q \geq 1/2$. The right half has no valley and dominates its reflection. A second peak occurs exactly when the density decreases somewhere on the left. Fix two points there. The value at the later point minus the value at the earlier point is affine in the bias and nonnegative at $q = 1$, by conditional unimodality. Once negative, it stays negative as the bias moves toward one half. Taking all such pairs gives an interval, and reflection gives its other half. Analyticity excludes flat pieces. We later use the two-mode bound again, this time to control the boundary ratio and prove continuity.

**Passing to the limit.** Suppose the limit density had a valley excluded by the finite bound. Take three short intervals around its high and low points. Their averages keep the strict inequalities, so weak convergence gives the same inequalities for the finite law. Choose the smallest atom in the middle interval and the largest in each outer interval. These three atoms give the forbidden finite valley. For two valleys we use five intervals instead (Lemma A.7). For curvature we first smooth the second differences with a Gaussian. Descartes' rule controls their signs. Summation by parts then gives the limit for a fixed Gaussian width, after which we let that width tend to zero.

**The unique transition.** Randomize the scale by an independent exponential variable. The moment equation of [8, (15)] then gives an explicit parametrization of the characteristic function. Two integrations by parts in its Mellin transform express $g_a''(0)$ as a weighted mean minus a constant. Rational bounds fix its sign outside $[2/3, 69/100]$. Inside that interval, increasing memory moves the weight to the right, so the sign crosses only once (Appendix B). To locate the zero and compute the critical derivatives, we separately integrate the moment series and bound its Fourier tail. The certificates and their error estimates are in Appendices C and D.

Section 2 gives the finite bounds and Section 3 their limiting consequences. Sections 4 and 5 record verification scope and open questions. Longer proofs are in the appendices.

### 1.4 Prior work

The counting chain predates the elephant walk. In the formulation of James, James and Qi [4, p. 2339], the Drezner–Farnum model [2] has next-success probability $(1 - \theta)p + \theta k/t$. Taking $p = 1/2$ and $\theta = a$ gives our transition rule. Drezner and Farnum's family includes bimodal laws. Heyde [3] established the fixed-parameter limit regimes, and James, James and Qi treat time-varying dependence. We classify limiting shapes as the initial bias and memory vary.

Fixed-memory results include [5, 6, 7]. Kiss and Vető [10] compute the first four limit moments for general step distributions. Peres and Qin [11] bound transition probabilities for step-reinforced walks on groups. Murahara and Tokumitsu [13] give optimal Gaussian local and central limit rates for fixed copying probabilities at most $3/4$. Vishwakarma [14] studies moment recurrences in the three regimes. We use the density and moments of [9, Theorems 1.3 and 1.4] and conditional unimodality from [8, Theorem 1.1]. The local

limit in Trinas and Valle's preprint [12, Theorem 3.1] could replace our weak-transfer argument. Their Theorem 4.4 disproves the log-concavity conjecture of [8, Remark 2.3].

The same transition appears in the growing-network model of Mohandas et al. [22]. They observe the symmetric bimodality transition and extrapolate its finite-length central curvatures to a copying probability near $0.836$ (Supplemental Material, Section IV.A). Our certified threshold is $(1 + a_*)/2 \approx 0.83440643$ in their parameter. The slow finite-length approach is visible in Remark 3.4.

Which mixing weights give two peaks is a classical question. Eisenberger [26] studies how the mixing weight affects bimodality for two normal components, and Robertson and Fryer [27] examine their shape across parameter space. Here the two components are a non-Gaussian limit density and its reflection. We prove that their bimodal weights form an interval by controlling the finite walk, without an explicit formula for the conditional density.

Dekking and Kong [20, Theorem 4.1] classify the Markov binomial law as unimodal, bimodal or trimodal. It depends on the previous state, so it differs from the walk here. Trinas [21] studies last returns to zero at fixed superdiffusive memory. His leading constant involves $g_{a,q}(0)$, which is independent of $q$ by reflection. The bias changes the shape away from zero even though it leaves that value unchanged.

The sign arguments use classical variation diminution, as developed in Karlin's work [24, 23]. For Gaussian smoothing we use the generalized Descartes rule from Pólya and Szegő [25, Part V, Chapter 1]. We include the elementary forms needed here. For this walk, we must also bound the boundary histories before applying the positive interior recurrence.

## 2 Finite shape bounds

The second difference of $P_N$ changes sign at most four times. We use this to count the limiting density's inflection points, and first differences to count its modes. Appendix A gives the first-difference argument for arbitrary initial bias and the passage to the limit.

For the fair-start second differences, $0 < \varepsilon < \frac{1}{2}$ and $a = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + ab/t$. For $t \geq 1$ and every integer $k$ we write $P_t(k) = \mathbb{P}(A_t = k)$, which is $0$ unless $0 \leq k \leq t$, and

$$D_t(k) = P_t(k+1) - 2P_t(k) + P_t(k-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \geq 2$.

**Lemma 2.1** (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer $k$,*

$$P_{t+1}(k) = \big(1 - r_t(k)\big)P_t(k) + r_t(k-1)\,P_t(k-1), \tag{2.1}$$

$$D_{t+1}(k) = \big(1 - r_t(k+1)\big)D_t(k) + r_t(k-2)\,D_t(k-1). \tag{2.2}$$

*Also $P_t(k) = P_t(t-k)$ and $D_t(k) = D_t(t-k)$.*

*Proof.* Given $A_t = k$, the copied step is $+1$ with probability $k/t$, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{k}{t} + \varepsilon(1-\frac{k}{t}) = r_t(k)$. This gives (2.1) for $0 \leq k \leq t+1$, where at $k = 0$ and at $k = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1-r_t(t+1))P_t(t+1)$, and this term is $0$ because $P_t(-1) = P_t(t+1) = 0$. For other $k$ every term vanishes. The symmetry follows by reversing every step, which preserves the law under a fair start. Taking second differences gives the symmetry of $D_t$.

For (2.2) put $\varsigma = a/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (2.1) at $k+1$, $k$ and $k-1$ into the definition of $D_{t+1}(k)$ and collect the coefficients of $P_t(k+1)$, $P_t(k)$, $P_t(k-1)$ and $P_t(k-2)$. They are

$$1 - r_t(k+1), \qquad 3r_t(k) - 2, \qquad 1 - 3r_t(k-1), \qquad r_t(k-2).$$

The right side of (2.2) has the coefficients $1 - r_t(k+1)$, $-2 + 2r_t(k+1) + r_t(k-2)$, $1 - r_t(k+1) - 2r_t(k-2)$ and $r_t(k-2)$. The second ones agree because $2r_t(k+1) + r_t(k-2) = 3r_t(k)$, and the third ones agree because $r_t(k+1) + 2r_t(k-2) = 3r_t(k-1)$. □

Trinas and Valle [12, (2.2)] use (2.1), the forward equation, for a superdiffusive local limit theorem. Their higher-difference recurrence [12, (4.10)] supports a certified fixed-memory study of log-concavity. We use (2.2) to count sign changes. Its coefficients are positive for $2 \le k \le t-1$, so new sign changes can enter only at the ends. The next lemma is the elementary variation-diminishing fact behind this argument (see [24, 23] for the general theory).

For a finite real list $z$, define its sign pattern $\mathrm{sp}(z)$ by deleting the zeros of $z$, replacing each remaining entry by its sign and merging equal neighbors. Thus $\mathrm{sp}(z)$ lists the alternating signs. For two such lists we write $p \preceq q$ if $p$ is a subsequence of $q$. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

**Lemma 2.2** (positive sums of neighbors)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix E.1.

**Proposition 2.3** (augmented rows and weak limits)**.** *Let laws on lattices of spacing $\eta_n \to 0$ converge weakly to a continuous density $g$. Include zero masses in each lattice row. Suppose that, for each $n$, its first differences occur in order in the last of a finite sequence of augmented rows. Each update has the form $z_i^{\mathrm{new}} = A_i z_i + B_i z_{i+1}$ with $A_i, B_i \ge 0$. If every initial row has sign pattern contained in $(+,-,+,-)$, then $g$ has no two successive strict valleys. If the same construction for differences on the positive half-line starts with pattern contained in $(+,-)$, then $g$ has no strict valley on $[0,\infty)$.*

*Proof.* Lemma 2.2 preserves the stated patterns through each finite sequence of rows. Two successive strict valleys would give differences with signs $-,+,-,+$, by telescoping. A single valley gives $-,+$. The respective patterns exclude these. Lemma A.7 transfers the exclusions to $g$ by averaging over short lattice windows. □

To apply this to the walk, we must construct the rows and check their initial signs. Figure 2 shows the construction for first differences. Proposition A.5 checks the full rows, and Proposition A.6 checks the half favored by the first step.

$$\underbrace{\ell_N, \dots, \ell_{t+1}}_{\text{left boundary history}} \qquad \underbrace{d_t(0), \dots, d_t(t-1)}_{\text{active row}} \qquad \underbrace{u_{t+1}, \dots, u_N}_{\text{right boundary history}}$$

$$\downarrow \text{ copy or sum adjacent entries}$$

$$\ell_N, \dots, \ell_{t+2} \qquad \ell_{t+1}, d_{t+1}(1), \dots, d_{t+1}(t-1), u_{t+1} \qquad u_{t+2}, \dots, u_N$$

Figure 2: Fix a final length $N$ and retain the boundary values that will enter later. Here $d_t(k) = P_t(k+1) - P_t(k)$, $\ell_t = d_t(0)$ and $u_t = d_t(t-1)$. The active row grows by one while the augmented row shrinks by one. Each incoming end is already in the history, so its sign is included in the preceding row's bound.

**Lemma 2.4** (sign changes next to an end)**.** *For every $t_1 \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \le \varepsilon \le 1$, the numbers $s_3$ and $s_4$ do not both vanish.*

We prove this in Appendix E.2.

**Theorem 2.5** (at most five blocks)**.** *Let $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\operatorname{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

*So the second difference of $P_N$ changes sign at most four times on $\{1, \dots, N-1\}$.*

We prove this in Appendix E.3.

## 3 The limit density at fixed memory

Fix $\varepsilon \in (0, \frac{1}{4})$, so $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. Conditional on $X_1 = 1$, $S_k/k^a$ converges almost surely to $\mathcal{L}_a$ [6, Theorem 3.7]. The fair-start limit $\mathcal{L}_a^\circ$ has the law of $\xi\mathcal{L}_a$, where $\xi$ is a fair sign independent of $\mathcal{L}_a$ [9, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [9, Theorem 1.3], denoted by $h_a$ and $g_a$ as before. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [8, Theorem 1.1].

### 3.1 The number of modes

A probability density $h$ on $\mathbb{R}$ is *unimodal* if for some $m$ it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [8], so we must also rule out a flat top. The density $g_a$ mixes the unimodal density of $\mathcal{L}_a$ with its reflection. Such a mixture can have more than one peak [17]. We first take a fair start. Lemma A.4 rules out constant intervals, so the modes are points. We consider other biases afterward.

**Theorem 3.1** (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

*(a) The density $g_a$ is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.*

*(b) There is $\kappa > 0$ such that $g_a$ is nondecreasing on $[0, \kappa]$ and nonincreasing on $[\kappa, \infty)$, and $g_a(0) < g_a(\kappa)$.*

*In case (b), $g_a$ is nondecreasing on $(-\infty, -\kappa]$ and nonincreasing on $[-\kappa, 0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is $0$ in case (a), and the only local maxima are $\pm\kappa$ in case (b). In case (a), there is $d > 0$ such that $g_a$ is strictly concave on $(-d, d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < \kappa < d_2$ such that $g_a$ is strictly convex on $(-d_1, d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2, -d_1)$ and $(d_1, d_2)$. Thus it has exactly two inflection points in case (a) and four in case (b).*

To pass from the second differences of $P_N$ to the curvature of $g_a$, we first smooth with a Gaussian. This gives an exponential sum to which Descartes' rule applies. We then let $N \to \infty$ at a fixed Gaussian width, and only afterward let the width tend to $0$. Appendix A proves the lemmas used below. A real function $v$ has *values at most* $\Pi$ if $\operatorname{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for all $y_1 < \cdots < y_j$, and *increments at most* $\Pi$ if $\operatorname{sp}(v(y_2) - v(y_1), \dots, v(y_j) - v(y_{j-1})) \preceq \Pi$ for all such points. We put $\Pi_4 = (+,-,+,-)$.

*Proof.* Fix $a$, so that $\varepsilon = (1-a)/2 \in (0, \frac{1}{4})$.

*Step 1, smoothing.* Fix $\sigma > 0$ and let $\varphi(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. For $N \geq 3$ put $\eta = 2N^{-a}$ and $z_k = (2k - N)N^{-a}$, so that $S_N/N^a = z_{A_N}$, and let

$$V_N(y) = \eta^{-2} \sum_{k=1}^{N-1} D_N(k)\, \varphi(y - z_k).$$

Since $\varphi(y - z_k) = \varphi(y) e^{-z_k^2/(2\sigma^2)} e^{y z_k/\sigma^2}$ and the $z_k$ increase with $k$, Lemma A.1 and Theorem 2.5 show that $V_N$ has values at most $\Pi_5$. (If every $D_N(k)$ vanishes, then $V_N = 0$ and every pattern is empty.)

*Step 2, the limit $N \to \infty$.* Let $u(y) = \mathbb{E}\varphi(y - \mathcal{L}_a^\circ)$. The derivatives of $\varphi$ are bounded, so $u$ is smooth and $u^{(j)}(y) = \mathbb{E}\varphi^{(j)}(y - \mathcal{L}_a^\circ)$. We show that $V_N(y) \to u''(y)$ for every $y$. Put $P_N(k) = 0$ for $k \notin \{0, \dots, N\}$. Then $D_N(k) = 0$ unless $-1 \le k \le N+1$, and summation by parts gives

$$\sum_{k=-1}^{N+1} D_N(k)\,\varphi(y - z_k) = \sum_{k=0}^{N} P_N(k)\,\Delta\varphi(y - z_k), \qquad \Delta\varphi(t) = \varphi(t+\eta) - 2\varphi(t) + \varphi(t-\eta).$$

Taylor's formula gives $|\eta^{-2}\Delta\varphi(t) - \varphi''(t)| \le \eta^2\|\varphi''''\|_\infty/12$. The four boundary terms on the left, with $k \in \{-1, 0, N, N+1\}$, have total absolute value at most $\varphi(0)\,(6P_N(0) + 2P_N(1))$, by symmetry. Here $P_N(0) = \frac{1}{2}(1-\varepsilon)^{N-1}$. The boundary recurrence gives $\xi_{n+1} = (1 - a/((1-\varepsilon)n))\xi_n + \varepsilon/(1-\varepsilon)$, where $\xi_n = P_n(1)/P_n(0)$ and $\xi_1 = 1$. Thus $\xi_n \le 1 + (n-1)\varepsilon/(1-\varepsilon)$ and $P_N(1) = O_\varepsilon(N(1-\varepsilon)^N)$. As $\eta^{-2} = N^{2a}/4$, these four terms times $\eta^{-2}$ tend to $0$. Hence

$$V_N(y) = \mathbb{E}\varphi''\big(y - S_N/N^a\big) + o(1) \longrightarrow \mathbb{E}\varphi''(y - \mathcal{L}_a^\circ) = u''(y),$$

because $S_N/N^a \to \mathcal{L}_a^\circ$ in law and $\varphi''$ is bounded and continuous. By Lemma A.2, $u''$ has values at most $\Pi_5$.

*Step 3, integrating the sign bound.* By dominated convergence, $u'(y) \to 0$ as $y \to \pm\infty$. So $u$ has increments at most $\Pi_4$, by Lemma A.3.

*Step 4, the limit $\sigma \to 0$.* Write $u = u_\sigma$. As $g_a$ is bounded and continuous, $u_\sigma(y) = \int g_a(y-t)\varphi(t)\,dt$ tends to $g_a(y)$ for every $y$. So the increments of $u_\sigma$ tend to those of $g_a$, and Lemma A.2 shows that $g_a$ has increments at most $\Pi_4$. Lemma A.4 also makes $g_a''$ bounded and continuous, so $u_\sigma'' = g_a'' * \varphi \to g_a''$ pointwise. Hence $g_a''$ has values at most $\Pi_5$.

*Step 5, symmetry.* Let $0 \le y_1 < \cdots < y_j$, let $d_i = g_a(y_{i+1}) - g_a(y_i)$, and let $p$ be the pattern of the $d_i$. As $g_a$ is even, the points $-y_j < \cdots < -y_1 < y_1 < \cdots < y_j$, with $-y_1$ left out if $y_1 = 0$, have the increments $-d_{j-1}, \dots, -d_1$, then $0$ if $y_1 > 0$, then $d_1, \dots, d_{j-1}$. Their pattern is $-p$ read backward, followed by $p$. It has $2|p|$ entries and is $\preceq \Pi_4$. So $p$ is empty, $(+)$, $(-)$ or $(+,-)$. In particular there are no $0 \le y < y' < y''$ with $g_a(y) > g_a(y') < g_a(y'')$.

If every increment on $[0,\infty)$ is nonpositive, case (a) holds. Otherwise choose $y_1 < y_2$ with $g_a(y_1) < g_a(y_2)$. Integrability supplies $y_3 < y_4$, both beyond $y_2$, with $g_a(y_3) > g_a(y_4)$. The sign bound excludes any later increase, so $g_a$ attains a maximum on $[0,\infty)$ at some $\kappa$. The absence of a strict valley makes it nondecreasing before $\kappa$ and nonincreasing afterward. The positive increment forces $\kappa > 0$. If $g_a(0) = g_a(\kappa)$, it would be constant on $[0,\kappa]$, contrary to analyticity and the positive increment. Thus $\kappa > 0$ and case (b) holds. Evenness gives the left-half shape and excludes unimodality in (b).

The monotonicities are strict because equality at two points would force a constant interval, contradicting Lemma A.4. This also excludes any additional local maximum.

*Step 6, curvature.* The Fourier bound in Lemma A.4 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the $\Pi_5$ bound give at most three sign blocks for $g_a''$ on $(0,\infty)$. Its last sign must be positive. Otherwise $g_a'$ would eventually be nonincreasing with limit zero, making $g_a$ eventually nondecreasing. This contradicts $g_a > 0$ and $g_a \to 0$. Analyticity and strict monotonicity imply that the first nonconstant term in the even Taylor series of $g_a$ at zero is negative in case (a) and positive in case (b). The same signs therefore hold for $g_a''$ near zero. In case (b), $g_a''$ must also be negative somewhere, since $g_a'$ is positive near zero and negative somewhere beyond $\kappa$. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < \kappa < d_2$. □

**Theorem 3.2** (two modes for memory near one)**.** *If $\frac{4}{5} \le a < 1$, that is $0 < \varepsilon \le \frac{1}{10}$, then case (b) of Theorem 3.1 holds. So $g_a$ has exactly two modes.*

*Proof.* In case (a), Khinchine's representation [17] gives $\mathcal{L}_a^\circ \overset{d}{=} UZ$, with $U$ uniform on $(0,1)$ and independent of $Z$. Hence Cauchy–Schwarz gives $\mathbb{E}(\mathcal{L}_a^\circ)^4 \ge \frac{9}{5}[\mathbb{E}(\mathcal{L}_a^\circ)^2]^2$. The moment recursion of [9, Theorem 1.4] instead gives

$$\frac{\mathbb{E}\mathcal{L}_a^4}{(\mathbb{E}\mathcal{L}_a^2)^2} = \frac{6(2a^2+2a-1)}{4a-1}\,\mathrm{Beta}(2a,2a),$$

where Beta is the beta function. For $a \ge \frac{3}{4}$, its integral and $t(1-t) \le \frac{1}{4}$ give $\mathrm{Beta}(2a,2a) \le 4^{-2a+3/2}\,\mathrm{Beta}(\frac{3}{2},\frac{3}{2}) = \pi 2^{-4a}$. Put $R(a) = (2a^2+2a-1)/(4a-1)$. On $[\frac{3}{4},1]$ we have $(\log R)' \le 2 < 4\log 2$. The first inequality reduces to $4a(4a^2+a-2) \ge 0$. Thus $6\pi R(a)2^{-4a}$ decreases there, and for $a \ge \frac{4}{5}$ the displayed moment ratio is at most

$$\frac{282\pi}{55\,2^{16/5}} < \frac{141}{80} < \frac{9}{5},$$

using $\pi < 22/7$ and $2^{1/5} > 8/7$. Even moments are unchanged by symmetrization, so this contradicts case (a). Theorem 3.1 then gives case (b). □

The moment argument gives an independent check near one. We now prove the complete transition by comparing the central curvature across memories.

**Theorem 3.3** (the unique transition)**.** *There is exactly one $a_* \in (1/2,1)$ with $g_{a_*}''(0) = 0$. It lies in $(2/3, 69/100)$, and*

$$g_a''(0) < 0 \quad (1/2 < a < a_*), \qquad g_a''(0) > 0 \quad (a_* < a < 1).$$

*Moreover, $\partial_a g_a''(0)|_{a=a_*} > 0$. The density $g_a$ is unimodal for $a \le a_*$ and bimodal for $a > a_*$. In particular, the critical density is unimodal.*

Appendix B proves the theorem without numerical input. The key step is to express the curvature sign as a comparison of positive weighted means. On the short interval where the sign can change, the rescaled weights have strictly increasing likelihood ratios as memory increases. The two endpoint signs then force exactly one crossing. Appendix C separately locates it near $0.6688$.

**Remark 3.4** (finite-length thresholds)**.** For the exact finite law, $D_N(N/2)$ changes sign at $a \approx 0.6563$ for $N = 4000$ and at $a \approx 0.6592$ for $N = 8000$. These computed examples are not used in any proof. They show how slowly the finite law approaches the limiting transition.

### 3.2 Arbitrary initial bias

Write $h_a$ for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

**Theorem 3.5** (the interval of bimodal biases)**.** *For every $a \in (1/2,1)$ and $q \in [0,1]$, the density $g_{a,q}$ has at most two local maxima. There is $b_a \in [0,1/2]$ such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

*Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$, in which case $b_a = 1/2$. At finite time, for every $0 < \varepsilon < 1$, $q \in [0,1]$ and $N \ge 1$, the count law has at most two modal plateaus, including its endpoints.*

*Proof. The mode bound.* Let $P_N$ here denote the count law with first-step bias $q$. The finite assertion follows from Proposition A.5. In particular, the count law cannot have two successive strict valleys. Neither can $g_{a,q}$. If it did, weak convergence of $S_N/N^a$ and averages over five short intervals would give five finite-law atoms with those same forbidden inequalities (Lemma A.7). There are also no constant intervals, for any bias, by Lemma A.4. Hence three local maxima $x < y < z$ would give $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair, take a one-sided neighborhood of the smaller maximum, toward the other one. Some point there has strictly smaller value, since the density is not constant on an interval. This gives the two valleys excluded by the limit bound.

*The bias interval.* Fix $a$, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition A.6 and the same weak-transfer lemma show that, for $q \geq 1/2$, $f_q$ has no strict valley on $[0, \infty)$ and $f_q(-x) \leq f_q(x)$ for $x \geq 0$. Consequently it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \leq 0$. Such a descent gives the valley $(x, y, -x)$ by reflection dominance. Conversely, a valley must start at $x < 0$. If its middle point $y$ is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \leq f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [8] excludes a valley, so $h$ is nondecreasing on $(-\infty, 0]$. For fixed $x < y \leq 0$, the difference $f_q(x) - f_q(y)$ is affine in $q$ and nonpositive at $q = 1$. If positive at $q \geq 1/2$, it remains positive for every $r \in [1/2, q]$. Thus, within $[1/2, 1]$, a valley persists when the bias decreases. The set of such biases is relatively open, since the strict inequalities at three fixed points survive a small change of bias. It excludes $1$. Reflection gives an open interval $(b_a, 1 - b_a)$, possibly empty.

A strict valley gives two local maxima. By integrability and positivity, we can choose an outer point on each side below the corresponding high point of the valley. Maximizing on the two resulting compact intervals gives the two local maxima. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 3.3 identifies the empty case. □

This proves the initial-bias conjecture of [8, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$. Trinas and Valle's stronger local limit [12, Theorem 3.1], in an unrefereed arXiv preprint, would also transfer the finite inequalities. Weak convergence suffices for the proof here.

### 3.3 The bias boundary

We find the endpoint by comparing the slopes of the conditional density at reflected points. At each pair, the ratio below tells us how much weight the reflected density needs to create a decrease.

**Proposition 3.6** (a variational equation for the bias boundary)**.** *For $a \in (1/2, 1)$ put $U_a = \{t > 0 : h_a'(t) > 0\}$. With the infimum over an empty set interpreted as $+\infty$,*

$$b_a = \min\left\{\frac{1}{2}, \inf_{t \in U_a} \frac{h_a'(-t)}{h_a'(t) + h_a'(-t)}\right\}. \tag{3.1}$$

*In particular $b_a \to 0$ as $a \uparrow 1$. If the infimum is attained at $t > 0$ with $h_a'(t) > 0$ and $b_a < 1/2$, then, with $q = 1 - b_a$,*

$$g_{a,q}'(-t) = g_{a,q}''(-t) = 0.$$

*At this point the smaller peak disappears.*

*Proof.* The proof of Theorem 3.5 shows that $h'_a(-t) \geq 0$ for $t \geq 0$. For $q \geq 1/2$ bimodality is equivalent to a decrease somewhere on the negative half-line, hence to $g'_{a,q}(-t) < 0$ for some $t > 0$. Here

$$g'_{a,q}(-t) = qh'_a(-t) - (1-q)h'_a(t).$$

This expression can be negative only if $t \in U_a$, in which case it is negative exactly when $1 - q > h'_a(-t)/(h'_a(t) + h'_a(-t))$. Taking the union over $t$ gives (3.1), with equality excluded as in the interval theorem. At an interior minimizer the ratio has derivative zero. Differentiating its numerator and denominator gives the two displayed equations. The moment estimate in Corollary 3.8 gives $b_a \to 0$. □

**Theorem 3.7** (regularity of the bias boundary)**.** *The function $a \mapsto b_a$ is continuous on $(1/2, 1)$, is continuously differentiable on $(a_*, 1)$, and extends continuously to $b_1 = 0$. For every $a > a_*$, we have $0 < b_a < 1/2$, and the infimum in (3.1) has a unique minimizer $t_a \in (0, \kappa_a)$. This minimizer depends continuously on $a$ throughout $(a_*, 1)$.*

*Proof.* For $a > a_*$, the curvature partition in Theorem 3.1 gives $g'_a(t) > 0$ on $(0, \kappa_a)$ and $g'_a(t) < 0$ on $(\kappa_a, \infty)$. For $t \geq \kappa_a$ in $U_a$, the ratio in (3.1) is therefore at least $1/2$. Since $b_a < 1/2$, it is enough to search $(0, \kappa_a)$ (we still have to show the infimum is attained). Put

$$D_a(t) = h'_a(t) + h'_a(-t), \qquad R_a(t) = \frac{g'_a(t)}{D_a(t)} \quad (0 < t < \kappa_a).$$

Lemmas A.8 and A.9 give $h'_a(-t) > 0$ for every $t \geq 0$. We therefore have $D_a > 0$, $0 < R_a < 1/2$, and $1/2 - b_a = \sup_{0<t<\kappa_a} R_a(t)$. Also $R_a(0+) = 0$.

We first rule out a strict valley in the ratio. We claim that

$$R_a(v) \geq \min\{R_a(u), R_a(w)\} \qquad (0 < u < v < w < \kappa_a). \tag{3.2}$$

Otherwise, choose $s \in (0, 1/2)$ strictly between the middle value and both outer values. With $q = 1/2 + s$, $g'_{a,q}(-t) = D_a(t)(s - R_a(t))$ then has signs $-, +, -$ at $-w, -v, -u$, and is positive sufficiently close to zero on the left. Positivity of the density and its zero limits at infinity supply a positive derivative farther left and a negative derivative farther right. There would be three local maxima, contrary to Theorem 3.5.

At $\kappa_a$, we have $D_a(\kappa_a) = 2h'_a(-\kappa_a) > 0$. Thus $R_a$ extends continuously to $[0, \kappa_a]$, with value zero at both endpoints. Its positive maximum is attained in the interior. At that point $h'_a(-t) > 0$, so the maximum is strictly below $1/2$. Hence $b_a > 0$. Attainment matters here, since pointwise strict inequality on an open interval would not give this conclusion. Equation (3.2) and analyticity make this maximum unique, since two maximizers would force a constant interval. They also make $R_a$ strictly increasing before its maximizer and strictly decreasing after it.

Lemma A.4 gives joint continuity of all the derivatives used here. The signs of $g'_a$ on either side of its unique positive zero give continuity of $\kappa_a$. Fix $a_0 > a_*$ and choose $0 < l < t_{a_0} < r < \kappa_{a_0}$. The strict inequalities $R_a(l), R_a(r) < R_a(t_{a_0})$ persist near $a_0$. For $v < l$, apply (3.2) to $v, l, t_{a_0}$. Since $R_a(l) < R_a(t_{a_0})$, it gives $R_a(v) \leq R_a(l)$. Applying it to $t_{a_0}, r, v$ gives $R_a(v) \leq R_a(r)$ for $v > r$. Thus every nearby global maximum lies in the fixed interval $[l, r]$. On this fixed compact interval the ratios converge uniformly, so their maximum values converge. Uniqueness forces their maximizers to converge too. This proves continuity above $a_*$. Below it $b_a = 1/2$, Proposition 3.10 gives continuity at $a_*$, and Proposition 3.6 gives the limit at one.

For differentiability, the same three-point argument puts all nearby maximizers in one compact interval. If $M(a) = 1/2 - b_a$, maximality gives, for $s > 0$,

$$R_{a+s}(t_a) - R_a(t_a) \leq M(a+s) - M(a) \leq R_{a+s}(t_{a+s}) - R_a(t_{a+s}).$$

Divide by $s$ and use the continuity of $t_a$ and $\partial_a R_a$ from Lemma A.4. The argument for $s < 0$ is the same. Thus

$$-b'_a = \frac{\partial_a h'_a(t_a)\, h'_a(-t_a) - h'_a(t_a)\, \partial_a h'_a(-t_a)}{\{h'_a(t_a) + h'_a(-t_a)\}^2}. \tag{3.3}$$

The right side is continuous. No derivative of $t_a$ is needed. □

The same argument gives a way to enclose the boundary. Verified inequalities $R_a(l), R_a(r) < R_a(t)$ with $0 < l < t < r < \kappa_a$ trap the maximum in $[l, r]$. If $L \le R_a(t)$ and $R_a \le U$ throughout that interval, then $1/2 - U \le b_a \le 1/2 - L$. Uniform inequalities give an enclosure for a whole parameter interval. For example, if $|R_a(s) - R_{a_0}(s)| \le M|a - a_0|$ for $s = l, t, r$, the two comparisons at $a_0$ hold throughout $|a - a_0| \le w$ whenever their gaps exceed $2Mw$. To prove strict decrease of $b_a$ on this interval, it would suffice to have

$$R_{a_2}(s) - R_{a_1}(s) \ge c(a_2 - a_1) \quad (l \le s \le r,\ a_1 < a_2), \qquad c > 0.$$

To see this, evaluate the later ratio at the earlier maximizer. We have not verified the required sign bound along the full curve.

**Corollary 3.8** (a rate for the bias boundary near one)**.** *Put*

$$\rho_a = \frac{\Gamma(1+a)^2}{(2a-1)\Gamma(2a)} - 1.$$

*If* $\rho_a < 1/12$*, then*

$$0 \le b_a \le \frac{4\rho_a}{1 - 4\rho_a}.$$

*In particular, as* $a \uparrow 1$*,* $b_a \le 8(1-a) + O((1-a)^2)$*.*

*Proof.* The conditional limit has mean $\mu_a = 1/\Gamma(1+a) > 0$ and variance $\rho_a \mu_a^2$, by the moment formulas of [9, Theorem 1.4]. Consider the three adjacent intervals of length $\mu_a$ centered at $-\mu_a$, $0$ and $\mu_a$. Chebyshev's inequality puts at least $1 - 4\rho_a$ of the conditional law in the right interval and at most $4\rho_a$ in the middle. For a start to the left, reflect these intervals. After mixing, the left interval has mass at least $(1-q)(1-4\rho_a)$ and the right at least $q(1-4\rho_a)$. The middle still has mass at most $4\rho_a$.

For $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$, the middle has less mass than either outer interval. All three intervals have the same length. So, by continuity, we can choose a point in each with the middle value strictly smaller than the other two. This rules out unimodality. The two-mode bound in Theorem 3.5 gives exactly two modes, and therefore the stated bound on $b_a$. Finally, the logarithmic derivative at $a = 1$ of $1 + \rho_a$ is $2\psi(2) - 2 - 2\psi(2) = -2$. Thus $\rho_a = 2(1-a) + O((1-a)^2)$, which gives the linear bound. □

### 3.4 The density at the threshold

How far apart are the two modes just after the transition? We compute this from derivatives at the threshold. Write

$$C_* = \partial_a g''_a(0)|_{a=a_*}, \qquad B_* = -g^{(4)}_{a_*}(0), \qquad H_* = h'_{a_*}(0).$$

Table 1 gives strict certified enclosures. The coefficient $Q_*$ is defined in Proposition 3.10.

| Constant | Lower bound | Upper bound |
|---|---|---|
| $C_*$ | 0.6889503945 | 0.6889503947 |
| $B_*$ | 0.079207784468 | 0.079207784470 |
| $H_*$ | 0.09707494510086 | 0.09707494510088 |
| $Q_*$ | 9.86698768 | 9.86698770 |
| $\sqrt{6C_*/B_*}$ | 7.224131934 | 7.224131936 |

Table 1: The constants in the critical expansions. The enclosure is uniform on a parameter interval containing the certified bracket for $a_*$ (Appendix D.3).

**Proposition 3.9** (the critical shape)**.** *The density at the threshold satisfies*

$$g_{a_*}(u) = g_{a_*}(0) - \frac{B_*}{24}u^4 + O(u^6) \qquad (u \to 0).$$

*For $a > a_*$ let $\kappa_a > 0$ be the positive mode of $g_a$. As $a \downarrow a_*$,*

$$\kappa_a = \sqrt{\frac{6C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big),$$
$$g_a(\kappa_a) - g_a(0) = \frac{3C_*^2}{2B_*}(a - a_*)^2 + o\big((a - a_*)^2\big).$$

*Proof.* On a neighborhood of $a_*$, the certificate bounds $|\phi_a(t)|$ by $Ct^{-8}$ and $|\partial_a\phi_a(t)|$ by $4t^{-7}$ for large $t$ (Appendix D.3). On a bounded frequency interval we instead use the moment series, which converges locally uniformly, as does its parameter derivative. These bounds let us pass limits and derivatives through the Fourier integral. The spatial derivatives through order six are continuous in $a$, and the central curvature has one parameter derivative. On a smaller parameter interval, the Taylor remainders through order four are uniform. At criticality, evenness, $g''_{a_*}(0) = 0$ and $g^{(4)}_{a_*}(0) = -B_* < 0$ give the first assertion.

Put $\delta = a - a_* > 0$. Uniform Taylor expansion gives

$$\frac{g'_a(u)}{u} = C_*\delta + o(\delta) - \frac{B_* + o(1)}{6}u^2 + O(u^4).$$

For $u = k\sqrt{\delta}$ this has the sign of $C_* - B_*k^2/6$ when $k$ is fixed away from $\sqrt{6C_*/B_*}$. The unique positive mode lies between any two such choices with opposite signs, by Theorem 3.1. Squeezing gives the mode expansion. Substitute it into

$$g_a(u) - g_a(0) = \tfrac{1}{2}\big(C_*\delta + o(\delta)\big)u^2 - \tfrac{1}{24}\big(B_* + o(1)\big)u^4 + O(u^6)$$

to obtain the height expansion. □

**Proposition 3.10** (how the bias interval opens)**.** *With the constants of Table 1, put*

$$Q_* = \frac{\sqrt{2}\,C_*^{3/2}}{3H_*\sqrt{B_*}}, \qquad Q_* \approx 9.86698769.$$

*As $a \downarrow a_*$,*

$$\frac{1}{2} - b_a = Q_*(a - a_*)^{3/2} + o\big((a - a_*)^{3/2}\big). \tag{3.4}$$

*On some interval immediately above $a_*$, $b_a$ is strictly decreasing. For $a > a_*$ sufficiently close to $a_*$, the infimum in (3.1) is attained. The minimizer $t_a$ of Theorem 3.7 satisfies*

$$t_a = \sqrt{\frac{2C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big) = \frac{\kappa_a}{\sqrt{3}} + o\big(\sqrt{a - a_*}\big).$$

*At $q = 1 - b_a$, the stationary point is $-t_a$.*

*Proof.* Proposition 3.9 supplies the required regularity. The refined certificate in Appendix D.3 encloses $\pi^{-1}\int_0^\infty t\,\mathrm{Im}\,\phi_a(t)\,dt = h'_a(0)$ uniformly on an interval containing $a_*$, giving the positive $H_*$ in Table 1.

Write $\delta = a - a_*$. By (3.1),

$$\frac{1}{2} - b_a = \max\left\{0, \sup_{t \in U_a} \frac{g_a'(t)}{h_a'(t) + h_a'(-t)}\right\}.$$

The fair-shape theorem makes the numerator nonpositive for $t \geq \kappa_a$. Proposition 3.9 gives $\kappa_a = O(\sqrt{\delta})$, and $H_* > 0$ makes both conditional derivatives positive on this shrinking interval. Uniform Taylor expansion gives, for bounded $v \geq 0$,

$$\frac{g_a'(v\sqrt{\delta})}{h_a'(v\sqrt{\delta}) + h_a'(-v\sqrt{\delta})} = \frac{\delta^{3/2}}{2H_*}\left(C_* v - \frac{B_* v^3}{6}\right) + o(\delta^{3/2}).$$

The cubic is largest at the unique positive point $v = \sqrt{2C_*/B_*}$. Maximizing on $[0, \kappa_a/\sqrt{\delta}]$, whose right endpoint tends to $\sqrt{6C_*/B_*}$, proves (3.4). Since the ratio vanishes at both endpoints, its positive maximum is attained inside $(0, \kappa_a)$. The convergence is uniform and the limiting cubic has a unique maximum, so every scaled maximizer tends to $\sqrt{2C_*/B_*}$. This proves the assertion about $t_a$. The bounds on $Q_*$ follow by squaring its formula and inserting the three rational enclosures for $C_*, B_*$ and $H_*$.

For strict monotonicity, choose a small fixed $r > 0$. For $a$ sufficiently close to $a_*$, the denominator of the ratio is positive on $[0, r]$ and $\kappa_a < r$. The maximum can therefore be taken over this fixed compact interval. The differentiated Fourier bounds justify the expansions $\partial_a g_a'(t) = C_* t + o(t)$ and $\partial_a(h_a'(t) + h_a'(-t)) = O(1)$, uniformly as $(a, t) \to (a_*, 0)$, while $g_a'(t) = O((|a - a_*| + t^2)t)$. The parameter derivative of the ratio is consequently positive for $0 < t \leq r$ after shrinking the parameter interval and $r$. At $a > a_*$ its maximum is positive and occurs at $t > 0$. Evaluating at that maximizer at any larger nearby memory makes the maximum strictly larger. Hence $b_a$ strictly decreases there. □

The two powers come from the width and height of the same cubic. Near zero the derivative is approximately $C_*(a - a_*)t - B_* t^3/6$. Its positive part has width of order $\sqrt{a - a_*}$ and height of order $(a - a_*)^{3/2}$. The width sets the distance between the modes. The height sets how much bias removes the smaller peak.

## 4 Verification

Table 2 separates analytic arguments from interval computations. The numerical illustrations are not used in any proof.

| Claim | How it is established | Trusted base |
|---|---|---|
| Finite shape bounds | Written recurrence and sign proofs | Exact finite identities and ordered sign patterns |
| Fair-limit shape, inflection counts, bias interval and boundary regularity | Written shape, transfer and maximization proofs | Cited limit law, conditional unimodality and Fourier representation |
| Unique memory threshold | Written positive-weight integral and likelihood-ratio proof | Moment equation, Fourier decay, Mellin continuation and rational bounds |
| Numerical threshold location | Endpoint sign certificates | Arb arithmetic and the proved moment and Fourier-tail bounds |
| Quartic critical top, emerging modes and bias width | Written Taylor and maximization arguments using certified derivatives | The same Fourier envelopes and interval enclosures |

Table 2: Proofs and their inputs. Appendix G gives the records needed to reproduce the computations.

The shared formalization uses Lean 4.33.1 and Mathlib at `0df444a360ea`, with only `propext`, `Classical.choice` and `Quot.sound` allowed by the axiom audit. Besides finite recurrences and reflection comparisons, it checks the analytic threshold proof's rational bounds, strict covariance argument, unique-crossing deductions and Mellin continuation. The actual $H_a$ series and its identification with the randomized characteristic function and curvature remain written proofs. The accompanying MAP.md records the hypotheses of every formal result and distinguishes current from archived results. Lean does not replay the transcendental interval enclosures. Boundary regularity and the critical expansions have written proofs.

The transition check retains its acceptance tests under `python -O`. A second check evaluates the signs at the two bracket endpoints directly, without the parameter Taylor models. It still uses Arb and the same analytic tail bound.

## 5 Open problems

**The full bias boundary.** Is $b_a$ strictly decreasing on $(a_*, 1)$? It decreases near $a_*$ (Proposition 3.10), and Figure 1 suggests the same elsewhere. By (3.3), it remains to determine a sign at the unique optimizer. We do not know that sign for all memories. The moment estimate gives $b_a \le 8(1-a) + O((1-a)^2)$. What is the exact decay near one?

**A finite-length threshold.** Does the central curvature have exactly one zero in $(0, 1)$ at every even length? The computations in Remark 3.4 suggest a threshold increasing toward $a_*$. Our limiting bias-interval theorem does not settle this finite-length question.

**The Drezner–Farnum model and higher dimensions.** Drezner and Farnum's family includes bimodal laws [2]. Changing its innovation probability $p$ changes the transition rule, while changing the first-step bias $q$ mixes two reflected laws. Our mixture argument therefore does not settle the shapes for $p \neq 1/2$. In higher dimensions, the fixed-point theory of [9] gives densities, but our sign argument needs points ordered along a line. Even the meaning of a central depression needs care (the usual axis-step walk is not rotationally symmetric). We have not computed either extension.

**Acknowledgments.** I thank Peter Kagey, whose Problem 131 of the Open Problem Collection [15] led to the crossing question, and Aidan Deshong for helpful conversations. The opening riddle recalls the crossing question that led to this work, namely to look for the first reversal.

## A note on the use of AI

I wrote the original draft of this paper and its mathematics. I used Claude Opus 5.5 (Anthropic, through Claude Code) for literature searches, citation checks and referee-style reviews that informed the framing and abstract. I read every source cited in the paper. I proposed and drafted the following additions: Lemma A.4; the strictness and inflection-point clauses of Theorem 3.1; the fourth-moment extension of Theorem 3.2 to $a \ge 4/5$; the Jensen lower bound in Theorem 4.5(a) of the archived combined version; and the weaker weak-convergence hypothesis in its Theorem 5.9(c). I used GPT-6 Astra (OpenAI, through Codex) to revise the manuscript and to revise the proofs of these additions. Proofs aside from the aforementioned were not altered by either model. The associated Lean files in kagey-problems/problem131/lean/PaperD were originally written by me and edited using GPT-6 Astra through Codex. Section 4 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take

responsibility for the claims, proofs and computations in the paper.

## Code and data

The tagged repository contains the source, certificate records and Lean files. The submission package also contains the numerical records and checksums. The archived combined version retains the rare-reversal results. For the assumptions and formal coverage, see Section 4.

## A Sign patterns and regularity

We prove the lemmas used in Theorem 3.1, with the notation of Sections 2 and 3.1. Recall that deleting entries can only remove signs from a list's pattern. An alternating list $p$ satisfies $p \preceq q$ for every alternating list $q$ with the same first sign and at least as many entries.

The following is the exponential-sum form of the generalized Descartes rule [25, Part V, Chapter 1]. We include the proof to keep track of the first sign as well as the number of changes.

**Lemma A.1** (Descartes' rule for exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_n$ and $b_1, \dots, b_n$ be real, with $b = (b_1, \dots, b_n) \neq 0$, and put $F(y) = \sum_{k=1}^{n} b_k e^{\lambda_k y}$. Then $F$ has finitely many real zeros, and $\operatorname{sp}(F(y_1), \dots, F(y_j)) \preceq \operatorname{sp}(b)$ for all $y_1 < \cdots < y_j$. The same holds for $\psi F$ for every positive function $\psi$.*

*Proof.* Let $Z(F)$ be the number of distinct real zeros of $F$, and $S(b)$ the number of sign changes of $b$. We show $Z(F) \leq S(b)$ by induction on $S(b)$. If $S(b) = 0$, then $F$ has the sign of the nonzero $b_k$ everywhere. If $S(b) \geq 1$, choose $i$ such that the last nonzero $b_k$ with $k \leq i$ and the first nonzero $b_k$ with $k > i$ have opposite signs, and choose $\lambda_i < \kappa < \lambda_{i+1}$. Then

$$G(y) = \frac{d}{dy}\big(e^{-\kappa y} F(y)\big) = \sum_{k=1}^{n} b_k (\lambda_k - \kappa) e^{(\lambda_k - \kappa) y}.$$

The coefficient $b_k(\lambda_k - \kappa)$ has the sign of $-b_k$ for $k \leq i$ and of $b_k$ for $k > i$, so the coefficients of $G$ have $S(b) - 1$ sign changes. By Rolle's theorem $G$ vanishes between any two zeros of $F$, so $Z(F) \leq Z(G) + 1 \leq S(b)$.

If $b_{k_0}$ is the first nonzero $b_k$, then $e^{-\lambda_{k_0} y} F(y) \to b_{k_0}$ as $y \to -\infty$. The zeros of $F$ cut the line into at most $S(b) + 1$ open intervals, on each of which $F$ has one sign. With equal neighbors merged, these signs form an alternating list that starts with the sign of $b_{k_0}$ and has at most $S(b) + 1$ entries, so it is $\preceq \operatorname{sp}(b)$. The nonzero values $F(y_1), \dots, F(y_j)$ lie in these intervals in order. A positive factor $\psi$ changes no sign. □

**Lemma A.2** (pointwise limits)**.** *Let $v_n$ and $v$ be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every $y$, and let $\Pi$ be an alternating list. If every $v_n$ has values at most $\Pi$, then so has $v$. If every $v_n$ has increments at most $\Pi$, then so has $v$.*

*Proof.* Let $y_1 < \cdots < y_j$, and let $I$ be the set of indices $i$ with $v(y_i) \neq 0$. For all large $n$, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \dots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i \in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \dots, v_n(y_j)) \preceq \Pi$. The increments of $v_n$ at $y_1, \dots, y_j$ tend to those of $v$, and the same argument applies to them. □

**Lemma A.3** (integrating the sign bound)**.** *Let $u$ be twice continuously differentiable on $\mathbb{R}$, with $u'(y) \to 0$ as $y \to \pm\infty$. If $u''$ has values at most $\Pi_5$, then $u'$ has values at most $\Pi_4$, and $u$ has increments at most $\Pi_4$.*

*Proof.* It is enough to treat points $y_1 < \cdots < y_j$ at which the values of $u'$ are nonzero and alternate in sign. Let $s$ be the sign of $u'(y_1)$. Choose $y_0 < y_1$ with $|u'(y_0)| < |u'(y_1)|$, and $y_{j+1} > y_j$ with $|u'(y_{j+1})| < |u'(y_j)|$. For $0 \le i \le j$ the mean value theorem gives $w_i \in (y_i, y_{i+1})$ with $u''(w_i)(y_{i+1} - y_i) = u'(y_{i+1}) - u'(y_i)$. The right side has the sign $s$ for $i = 0$, the sign of $u'(y_{i+1})$ for $1 \le i \le j-1$, and the sign of $-u'(y_j)$ for $i = j$. In each case this is $(-1)^i s$. So $u''$ takes $j+1$ alternating signs, starting with $s$, at $w_0 < \cdots < w_j$. This list is $\preceq \Pi_5$, so $j + 1 \le 5$, and $j + 1 \le 4$ if $s$ is $-$. In both cases the list of the $j$ signs of $u'(y_1), \ldots, u'(y_j)$ is $\preceq \Pi_4$.

For the increments at $y_1 < \cdots < y_j$, each nonzero $u(y_{i+1}) - u(y_i)$ has the sign of $u'(\xi_i)$ for some $\xi_i \in (y_i, y_{i+1})$, by the mean value theorem. So the pattern of the increments is a pattern of values of $u'$. □

**Lemma A.4** (entire densities and parameter continuity)**.** *For every $a \in (\frac{1}{2}, 1)$ and $1 < r < 1/a$, the characteristic function $\phi_a$ of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)| \le 2e^{-c|t|^r} \qquad (t \in \mathbb{R})$$

*for some $c = c(a, r) > 0$. The same $c$ works on a sufficiently small neighborhood of any fixed $a$. Every spatial derivative of $h_a$ and its first derivative in $a$ are jointly continuous in $(a, u)$. For every $q \in [0, 1]$, the density of the limit with first-step bias $q$ extends to an entire function and is not constant on an interval of positive length.*

*Proof.* First, we check weak continuity in $a$. On a compact $J \subset (1/2, 1)$, the martingale $\mathcal{M}_n(a) = \Gamma(n)S_n/\Gamma(n + a)$ of [6, (2.4), Appendix A] satisfies

$$\sup_{a \in J} \mathbb{E}|\mathcal{L}_a - \mathcal{M}_n(a)|^2 \le C_J \sum_{k \ge n} k^{-2 \min J} \longrightarrow 0.$$

The walk's centered next step has conditional variance at most one, and the gamma ratios are uniformly bounded on $J$. The finite laws are continuous in $a$, so $\phi_a(t)$ is jointly continuous, uniformly on compact frequency intervals.

The fixed-point equation of [9, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)| \le \int_0^1 |\phi_a(tv^a)|\, |\phi_a(t(1-v)^a)|\, dv.$$

Here $\phi_a$ is the characteristic function conditional on a positive first step. The random sign disappears after taking absolute values, since $|\phi_a(-s)| = |\phi_a(s)|$. Fix $a_0$ and $1 < r < 1/a_0$, and choose $J$ about $a_0$ with $r \max J < 1$. Put $D(v) = v^{r \max J} + (1 - v)^{r \max J} - 1$. We have $D(v) > 0$ for $0 < v < 1$. Dominated convergence therefore permits $K > 1$ such that

$$R := 2 \int_0^1 e^{-K^r (\log 2) D(v)}\, dv < 1.$$

The law of $\mathcal{L}_{a_0}$ has a density [9, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T > 0$ with $|\phi_{a_0}(t)| < 2^{-K^r}$ for $|t| \ge T$. Joint continuity, after shrinking $J$, gives $|\phi_a(t)| \le 2^{1-K^r}$ for $a \in J$ and $T \le |t| \le KT$. Set $c = (\log 2)/T^r$ and $b(t) = 2e^{-ct^r}$ for $t \ge 0$. On $[0, T]$ we have $b \ge 1$, and on $[T, KT]$ we have $b \ge 2^{1-K^r}$. Thus $|\phi_a| \le b$ on $[0, KT]$.

For $t \ge KT$,

$$\frac{\int_0^1 b(tv^a) b(t(1-v)^a)\, dv}{b(t)} \le 2 \int_0^1 e^{-ct^r D(v)}\, dv \le R < 1.$$

Suppose the bound $|\phi_a| \le b$ stops at a finite $t_* \ge KT$. By continuity it still holds on $[0, t_*]$. At the endpoint the fixed-point inequality gives the strict bound $|\phi_a(t_*)| \le Rb(t_*) < b(t_*)$.

Continuity then extends the interval past $t_*$, a contradiction. The bound thus holds on $[0,\infty)$, and evenness extends it to $\mathbb{R}$.

Since $r > 1$, the Fourier integral converges absolutely and locally uniformly for complex arguments. Indeed, for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|}|\phi_a(t)|\,dt < \infty$. The Fourier integral therefore defines an entire function. After any fixed number of spatial derivatives, the bound on $J$ still supplies an integrable majorant, giving joint continuity by dominated convergence. For initial bias $q$, apply the triangle inequality to $q\phi_a(t) + (1-q)\phi_a(-t)$ to get the same bound. A density of integral one cannot be constant on $\mathbb{R}$. If it were constant on an interval, the identity theorem would make it constant everywhere.

For the memory derivative, Lemma D.1(b) gives a complex moment majorant and local parameter differentiation on every compact frequency interval. The differentiated smoothing inequalities in Lemma D.8 extend this to Fourier inversion. Indeed, for any prescribed power $\beta$, the exponential bound above supplies $f(t) \le Ct^{-\beta}$. Taking $T$ large makes both contraction coefficients $R$ and $\kappa$ in that lemma smaller than one. Choose $D_1$ to bound the frequency derivative on $[T_2, T]$ and to satisfy its strict contraction inequality, then choose $D_a$ in the same way for the memory derivative. The maximum argument in that proof gives $|\partial_a \phi_a(t)| = O_J((1+|t|)^{1-\beta})$. Here $\beta$ is arbitrary and the compact memory interval can be shrunk about any fixed parameter. Multiplication by any fixed power of $t$ therefore leaves an integrable majorant. Differentiation under Fourier inversion proves the asserted joint continuity. □

### A.1 First differences for a biased start

**Proposition A.5** (two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0,1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \ge 1$ steps. Extend $P_n$ by zero outside $\{0,\dots,n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\operatorname{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+,-,+,-).$$

*In particular, $P_n$ has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which*

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

*Proof.* Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \le k \le t-1. \tag{A.1}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$, since $r_t(j)$ is a convex combination of $\varepsilon$ and $1-\varepsilon$ for $0 \le j \le t$. It remains to check the two ends.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{A.2}$$

Once an increment is nonnegative, every later increment is positive. Thus $\xi_t$ first decreases and then increases (either part may be empty). For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (+, -, +).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (-, +)$. In particular, a positive $\ell_i$ cannot be followed by a negative $\ell_j$. Zeros do not affect either conclusion.

Write $u_t = d_t(t-1)$. Reflection of the walk gives $u_t(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-, +, -)$. For $q = 0$ it is nonpositive throughout. The conditional count is at most $t - 1$, so $u_t = -P_t(t-1) \le 0$.

We now keep the boundary values that the later rows will need. Fix $N$ and, for $1 \le t \le N$, form the augmented row

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), u_{t+1}, \dots, u_N).$$

Read the left history backward and the right history forward. Each incoming endpoint is then already in the preceding augmented row. After checking the two histories, we can therefore use the interior recurrence to control the whole row. For $q \le 1/2$, its first row is contained in $(+, -, +, -)$. We must also allow zeros at $\ell_1 = u_1 = 2q - 1$. Suppose $Z_1$ contained four ordered signs $-, +, -, +$. If the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+, -, +$. For $q > 0$, their reflected negatives give the forbidden pattern $(-, +, -)$ in the left history at bias $1 - q$. For $q = 0$, they contradict the nonpositive right history. This also covers $N = 1$.

Every entry of $Z_{t+1}$ is a nonnegative combination of adjacent entries of $Z_t$. The interior entries use (A.1). The two new boundary entries and the remaining future edge values are copied, with coefficients $(1, 0)$ or $(0, 1)$. To check the ordering, number the augmented entries from left to right. Output entry $i$ uses input entries $i$ and $i + 1$, including copied entries by allowing one coefficient to vanish. A nonzero output has a parent of the same sign. Two consecutive opposite signs cannot choose the same parent (one entry cannot have both signs), so their parent indices strictly increase. Choosing parents this way for any alternating output subsequence gives the same signs in order in the input. An update therefore cannot add an alternating pattern, so $\operatorname{sp}(Z_N) \preceq (+, -, +, -)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \ge 0$ on the left and $d_N(N) = -P_N(N) \le 0$ on the right. Neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the last displayed inequalities in the statement held, each of their four intervals would contain an adjacent difference of the same strict sign, by telescoping. These four differences would have signs $-, +, -, +$, a contradiction. □

**Proposition A.6** (the dominant half)**.** *Under the hypotheses of Proposition A.5, suppose $q \ge 1/2$. Then $P_n(n-k) \le P_n(k)$ for $2k \ge n$, and the right half has no strict valley. That is,*

$$P_n(j) \ge \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i + 1 \ge n.$$

*Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \le 1/2$.*

*Proof.* Write the bias explicitly as $P_{n,q}$ and put

$$A_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $A_{n+1}(k) = (1 - r_n(k))A_n(k) + r_n(k-1)A_n(k-1)$. At $n = 1$, $A_1(1) = 2q - 1 \geq 0$. Suppose the assertion holds at length $n$. At a point strictly above the new center, both parents lie in the preceding upper half and both coefficients are nonnegative. At an integer center $A_{n+1}(k) = 0$ by reflection. Terms outside the support vanish. This proves the induction. In particular, the central first difference of an odd row is nonnegative.

Restrict the augmented row $Z_t$ in the preceding proof to spatial indices $2k + 1 \geq t$. At $t = 1$ its pattern is contained in $(+, -)$. The central entry is $2q - 1 \geq 0$, followed by the future right boundary values $u_s(q) = -\ell_s(1 - q)$. Since $1 - q \leq 1/2$, their pattern is $(+, -)$ by (A.2). We now rule out a negative difference followed by a positive one, also among the future boundary entries. Suppose such a pair first appeared in the next row. Choose a same-sign parent for each entry. The preceding proof shows that the first parent comes before the second. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index $i$ of row $t + 1$ satisfies $2i + 1 \geq t + 1$, then $2i > t$, so both its parents satisfy $2k + 1 \geq t$. Parents of a later positive entry follow them strictly, as in the preceding proof. Induction therefore excludes the pair in every restricted augmented row. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

**Lemma A.7** (weak transfer of shape)**.** *Let probability laws on lattices $s_n + \eta_n\mathbb{Z}$, with $\eta_n > 0$ and $\eta_n \to 0$, converge weakly to a law with continuous density $g$. If their lattice mass sequences, including zero masses, exclude two successive strict valleys, then so does $g$. If they exclude a strict valley among positive lattice points, then $g$ has no strict valley on $[0, \infty)$. For reflection-symmetric lattices, if the mass at each positive lattice point is at least that at its reflection, then $g(x) \geq g(-x)$ for $x \geq 0$.*

*Proof.* For a bounded interval $J$ of positive length, let $K_n(J)$ be its lattice indices and let $\mu_n(J)$ be its probability. Lattice counting gives $|\eta_n \# K_n(J) - |J|| \leq 2\eta_n$, so these sets are eventually nonempty. Since the limiting law has no atoms, weak convergence gives

$$B_n(J) := \frac{\mu_n(J)}{\eta_n \# K_n(J)} \longrightarrow \frac{1}{|J|} \int_J g(x)\, dx.$$

Thus $B_n(J)$ is the average scaled atom mass on $J$. If $g$ had two successive strict valleys, take five disjoint equal-length intervals around their high and low points. Since $g$ is continuous, we can make them short enough that every value on a high interval exceeds every value on either neighboring low interval (there are only four comparisons to preserve). Their averages therefore retain all four inequalities. The corresponding $B_n$ satisfy these inequalities for large $n$. Choose the largest mass in the first, third and fifth intervals, and the smallest in each of the other two. These points are in order because the intervals are disjoint and ordered. Each largest mass is at least its interval average, and each smallest mass is at most its average, so all four strict comparisons hold for these atoms. We include zero masses in this selection (a low interval might miss the finite support). The selected atoms give the forbidden two valleys.

For one valley on the positive half-line, use three intervals and the same choice of largest, smallest and largest mass. If the first witness is $0$, continuity moves it slightly to the right while preserving the strict inequalities. Finally, for any interval $J \subset (0, \infty)$ the reflection hypothesis gives $\mu_n(-J) \leq \mu_n(J)$. Weak convergence passes this inequality to

the limiting interval masses. Shrink $J$ to $x > 0$ and use continuity to get $g(-x) \le g(x)$. At $x = 0$ there is equality. □

### A.2 A start with one step in each direction

**Lemma A.8.** *Fix the first two steps to be $+1, -1$. For $0 \le a \le 1$, the count law at each length $n \ge 2$ is log-concave, with positive support $\{1, \dots, n-1\}$. If $1/2 < a < 1$, its limiting density $f_a^{+-}$ is symmetric and log-concave, with $(f_a^{+-})'(t) < 0$ for $t > 0$. Consequently the density conditional on a positive first step satisfies*

$$h_a'(t) > 0 \qquad (t < 0).$$

*Proof.* With one initial step in each direction, the boundary terms no longer obstruct the log-concavity induction of [8, Proposition 2.2]. Write one update as $g_k = Af_k + Bf_{k-1}$, where

$$A = \frac{1+a}{2} - \frac{ak}{n}, \qquad B = \frac{1-a}{2} + \frac{a(k-1)}{n}, \qquad d = \frac{a}{n}.$$

For $u = f_{k-2}$, $v = f_{k-1}$, $w = f_k$ and $z = f_{k+1}$, expansion gives

$$\begin{aligned} g_k^2 - g_{k-1}g_{k+1} = (A^2 - d^2)(w^2 - vz) + (B^2 - d^2)(v^2 - uw) \\ + (A - d)(B - d)(wv - uz) + d^2(w - v)^2. \end{aligned}$$

For $2 \le k \le n-1$ we have $A, B \ge d \ge 0$. The preceding row is log-concave with interval support, so all three gaps are nonnegative. At either boundary an adjacent output mass vanishes. Starting from the point mass at one now proves log-concavity by induction. The same recurrence gives strict positivity throughout the support.

Log-linear interpolation of these lattice masses is log-concave. Its integral differs from that of the step interpolation by at most twice the largest atom, which tends to zero because the limiting law has a continuous density. Thus the limit is log-concave. The two-subtree representation gives

$$\mathcal{L}_a^{+-} \overset{d}{=} U^a \mathcal{L}_a' - (1-U)^a \mathcal{L}_a'',$$

with independent copies and an independent uniform $U$. This proves symmetry. It also gives positivity and an entire density, by positivity of $h_a$ and the Fourier bound of Lemma A.4. The derivative of $\log f_a^{+-}$ is nonincreasing and vanishes at zero. If it vanished at a positive point, the density would be constant on an interval, contradicting analyticity. Hence $(f_a^{+-})' < 0$ on the positive half-line.

For two positive initial steps, reflection dominance and the first-difference recurrence give nonnegative first differences on the negative half-line. The induction starts from the point mass at count two. Both parents stay in the preceding lower half except at the central difference of an odd row, where reflection dominance gives the required sign. Passing to the derivative limit as in Lemma F.1 gives $(f_a^{++})'(t) \ge 0$ for $t < 0$. Finally $h_a = ((1+a)/2)f_a^{++} + ((1-a)/2)f_a^{+-}$. The second summand has strictly positive derivative there. □

**Lemma A.9** (the conditional mode is positive)**.** *For every $1/2 < a < 1$, we have $h_a'(0) > 0$. In particular the unique conditional mode lies strictly to the right of zero.*

*Proof.* We need a lower bound on the reflection difference that stays positive after rescaling and passing to the limit. Write $f_n(k) = \mathbb{P}_+(A_n = k)$ and, on $z \equiv n \pmod 2$, put $d_n(z) = f_n((n+z)/2) - f_n((n-z)/2)$. The reflection induction in Proposition A.6 is strict

for $0 < z \le n$ when the start is positive. Every noncentral upper-half entry has a positive parent with positive coefficient, starting from $d_1(1) = 1$. The recurrence is

$$d_{n+1}(z) = A_n(z)d_n(z+1) + B_n(z)d_n(z-1),$$
$$A_n(z) = \frac{1}{2} - \frac{a(z+1)}{2n}, \qquad B_n(z) = \frac{1}{2} + \frac{a(z-1)}{2n}.$$

Fix $n_0 > 4a$ and divide by $z$. After multiplication by $\prod_{j=n_0}^{n-1}(1-2a/j)^{-1}$, the resulting values $v_n(z)$ satisfy a backward averaging equation with weights

$$q_+ = \frac{A_n(z)(z+1)}{z(1-2a/n)}, \qquad q_- = \frac{B_n(z)(z-1)}{z(1-2a/n)}.$$

The weights sum to one and are nonnegative in the interior band used below. At $z = 1$, the weight of the parent zero vanishes. For the corresponding backward chain,

$$\mathbb{E}[z_n^2 \mid z_{n+1} = z] = \frac{n-4a}{n-2a}z^2 + \frac{3n-4a}{n-2a}.$$

Set $\gamma_n = \Gamma(n-2a)/\Gamma(n-4a) \sim n^{2a}$. Then $Y_n = z_n^2/\gamma_n$ has backward drift $\delta_n = (3n-4a)/\{(n-2a)\gamma_n\}$, and $\sum_{n\ge n_0}\delta_n \to 0$ as $n_0 \to \infty$.

Fix $R > 0$. Start at $z_N = 1$ for odd $N$ and $z_N = 2$ for even $N$, and stop at row $n_0$ or on leaving $0 < z/\sqrt{\gamma_n} \le R$. For sufficiently large $n_0$, every band state and its immediate parents lie inside the support. The expected stopped value of $Y$ is at most $z_N^2/\gamma_N + \sum_{n\ge n_0}\delta_n$. At an exit it exceeds $R^2$, so increasing $n_0$ and then $N$ makes the exit probability less than one half. All stopped $v$ values are positive. At row $n_0$ there are only finitely many band states, so their minimum $m_0$ is positive. The chain reaches that row with probability greater than one half. The bounded stopping argument therefore gives $v_N(z_N) \ge m_0/2$. The product normalizer is asymptotic to a positive constant times $N^{2a}$, and hence

$$\liminf_N N^{2a}\frac{d_N(z_N)}{z_N} > 0.$$

The first-difference convergence proved in Lemma F.1, applied to the conditional start, identifies the limit as $4h_a'(0)$. Conditional unimodality then puts its mode to the right of zero. □

## B An analytic proof of the transition

To prove Theorem 3.3, we use the moment equation of Guérin, Laulin, Raschel and Simon [8, (15)] to write the central curvature as an integral. Its weights are positive, so we can compare them as memory varies.

Put

$$\chi_k = 4^{-k}\binom{2k}{k}, \qquad H_a(y) = 1 - \sum_{k\ge1}\frac{\chi_k}{2ak-1}y^k. \tag{B.1}$$

The function $H_a$ decreases strictly on $[0,1)$. For $a = 1$ it is $\sqrt{1-y}$. For $a < 1$ every subtracted coefficient is larger, so $H_a(1) < 0$. The series at one converges, since its coefficients are bounded by a constant depending on $a$ times $\chi_k/(2k-1)$, whose sum is one. There is consequently a unique root $Y_a \in (0,1)$. The coefficients also give

$$H_a(y) - 2ayH_a'(y) = (1-y)^{-1/2}. \tag{B.2}$$

Since $\partial_a H_a(y) > 0$ for $y > 0$ and $H_a'(Y_a) < 0$, the root is smooth and $Y_a' > 0$.

**Lemma B.1.** *For $1/2 < a < 1$,*

$$g_a''(0) = -\frac{1}{\pi\Gamma(3-3a)} \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a} \big[1 - (1+a)^2 y\big]\, dy. \tag{B.3}$$

*The integral is absolutely convergent.*

*Proof.* We first multiply the limit by an independent exponential power (this cancels the gamma factors in its moments). Let $E$ be an independent exponential random variable of mean one and set $Z_a = E^a \mathcal{L}_a$. With the normalization of [8],

$$m_n = \frac{\Gamma(1+an)}{n!} \mathbb{E}\mathcal{L}_a^n, \qquad m_0 = m_1 = 1.$$

The bound $m_n \le C_a^n$ in [8, Section 3], applied to the even moments, gives $\mathbb{E}\exp(\delta|Z_a|) < \infty$ for small $\delta > 0$. Hence $M_a(z) = \mathbb{E}e^{zZ_a}$ is analytic on a strip about the imaginary axis. Its power series and the moment equation give

$$M_a(z) + azM_a'(z) = \frac{1+a}{2} M_a(z)^2 + \frac{1-a}{2} M_a(z) M_a(-z). \tag{B.4}$$

The identity holds on the strip by analytic continuation. If $M_a(it) = A(t) + iB(t)$, it becomes

$$atA' = A^2 - A - aB^2, \qquad atB' = ((1+a)A - 1)B. \tag{B.5}$$

For $0 < y < Y_a$ define

$$t = \sqrt{y}\, H_a(y)^{-a}, \qquad A(t) = H_a(y)\sqrt{1-y}, \qquad B(t) = H_a(y)\sqrt{y}. \tag{B.6}$$

The map from $y$ to $t$ increases from zero to infinity. Equation (B.2) gives

$$\frac{1}{t}\frac{dt}{dy} = \frac{1}{2yH_a(y)\sqrt{1-y}},$$

and substitution shows that (B.6) satisfies (B.5). Near zero, $A$ is even, $B$ is odd, and both are analytic in $t$, with $A(0) = 1$ and $B'(0) = 1$. These data determine its power series. Indeed, the coefficient recurrence has denominator $na - 1$ for even $n \ge 2$, and $na - a$ for odd $n \ge 3$, both positive. Thus it agrees locally with $M_a(it)$. Ordinary uniqueness for (B.5) on $t > 0$ extends the agreement to every positive $t$.

Write $f(t) = \operatorname{Re}\mathbb{E}e^{it\mathcal{L}_a}$. Exponential randomization gives

$$A(t) = \int_0^\infty e^{-v} f(tv^a)\, dv.$$

We use the Fourier decay in Lemma A.4, which in particular makes every polynomial multiple of $f$ integrable at infinity. For $0 < \Re s < 1/a$, absolute Fubini gives

$$J(s) := \int_0^\infty t^{s-1} A(t)\, dt = \Gamma(1-as)\mathcal{F}(s), \qquad \mathcal{F}(s) = \int_0^\infty t^{s-1} f(t)\, dt. \tag{B.7}$$

The function $\mathcal{F}$ is holomorphic on $\Re s > 0$. The integral defining $J$ diverges at $s = 3$, which is the value we need. Two integrations by parts give a convergent expression there.

Set $\theta = \arcsin\sqrt{y}$ and

$$u = F(\theta) = \sin^{-1/a}\theta\, H_a(\sin^2\theta), \qquad w(u) = a\cos\theta \sin^{1/a}\theta.$$

Here $\theta$ decreases from $\arcsin\sqrt{Y_a}$ to zero as $u$ increases from zero to infinity. From (B.2),

$$F'(\theta) = -\frac{1}{a}\sin^{-1-1/a}\theta, \qquad t = u^{-a}.$$

Changing variables in $J$ gives $J(s) = \int_0^\infty u^{-as}w(u)\,du$. Since $F'(\arcsin\sqrt{Y_a}) \neq 0$, the function $w$ is smooth at zero. At infinity, $\theta \sim u^{-a}$, so its definition and the derivative formulas below give $w(u) = O(u^{-1})$, $w'(u) = O(u^{-2})$ and $w''(u) = O(u^{-3})$. For $0 < \Re s < 1/a$, the boundary terms therefore vanish twice, and (B.7) becomes

$$\mathcal{F}(s) = \frac{1}{\Gamma(3-as)}\int_0^\infty u^{2-as}w''(u)\,du. \tag{B.8}$$

The right side is holomorphic on $0 < \Re s < 3/a$, using the stated bounds at infinity and boundedness of $w''$ at zero. Thus (B.8) holds throughout this larger strip by the identity theorem, and $s = 3$ is permitted.

Differentiating through $u = F(\theta)$ gives

$$\begin{aligned} w'(u) &= -a\sin^{2/a}\theta[1-(1+a)\sin^2\theta], \\ w''(u) &= 2a\sin^{3/a}\theta\cos\theta[1-(1+a)^2\sin^2\theta]. \end{aligned}$$

Insert the second expression in (B.8) at $s = 3$ and return to $y = \sin^2\theta$. The powers of $y$ reduce to $y^{1/2}$, giving the integral in (B.3). Fourier inversion supplies the factor $g_a''(0) = -\mathcal{F}(3)/\pi$. Finally, $H_a$ has a simple zero at $Y_a$, and $2 - 3a > -1$, so the endpoint singularity is integrable. □

*Proof of Theorem 3.3. Signs outside a short interval.*

Let $\mu_a$ be the probability law on $(0, Y_a)$ with density proportional to $y^{1/2}H_a(y)^{2-3a}$, and put

$$\mathcal{R}(a) = (1+a)^2\int y\,d\mu_a(y). \tag{B.9}$$

Lemma B.1 says that $g_a''(0)$ has the sign of $\mathcal{R}(a) - 1$.

When $a \le 2/3$, the extra weight $H_a^{2-3a}$ decreases in $y$. It therefore lowers the mean of the density proportional to $y^{1/2}$, giving $\int y\,d\mu_a(y) \le 3Y_a/5$. At $a = 2/3$, the first three terms of (B.1) give

$$H_{2/3}(3/5) \le 1 - \frac{9}{10} - \frac{81}{1000} - \frac{9}{400} = -\frac{7}{2000} < 0.$$

Since $Y_a$ increases, $Y_a \le Y_{2/3} < 3/5$ for $a \le 2/3$. Thus $\mathcal{R}(a) < (5/3)^2(3/5)^2 = 1$.

When $a \ge 2/3$, the extra weight instead increases, so the mean is at least $3Y_a/5$. We need only the lower bound $Y_{2/3} > 117/200$. To prove it, note that $\chi_k$ decreases. At $y = 117/200$, the terms with $k \ge 4$ are at most

$$\sum_{k\ge 4}\frac{\chi_k y^k}{4k/3-1} \le \frac{\chi_4 y^4}{(13/3)(1-y)}.$$

Together with the first three terms this gives the exact bound

$$H_{2/3}(117/200) \ge \frac{929771}{135987200} > 0.$$

For every $a \ge 69/100$, therefore,

$$\mathcal{R}(a) > \left(\frac{169}{100}\right)^2\frac{3}{5}\frac{117}{200} = \frac{10024911}{10000000} > 1.$$

Any zero must lie between $2/3$ and $69/100$.

*A single crossing.*

We now work on $I = [2/3, 69/100]$. Rescale the integration interval by $y = Y_a u$, so the variable $u$ always lies in $(0, 1)$. Put

$$w_k(a) = \frac{\chi_k Y_a^k}{2ak-1}, \qquad K_a(u) = \frac{H_a(Y_a u)}{1-u} = \sum_{k\geq 1} w_k(a)(1 + u + \cdots + u^{k-1}). \tag{B.10}$$

Since $\sum w_k = 1$, we have $K_a \geq 1$ and $\partial_u K_a \geq 0$. Writing $s_a = 3a - 2 \in [0, 7/100]$, the rescaled measure has density proportional to

$$W_a(u) = u^{1/2}(1-u)^{-s_a} K_a(u)^{-s_a}. \tag{B.11}$$

We will show that its density ratios increase strictly in $u$ as $a$ increases (so the mean increases too).

First, $Y_a < 2/3$ throughout $I$. Indeed, the first three terms give

$$H_{69/100}(2/3) \leq -\frac{9767}{7087608} < 0,$$

and $Y_a$ increases. All series and derivatives used below therefore converge uniformly on $I \times [0, 1]$. Define the falling-factorial moments

$$M_1 = \sum k w_k, \qquad M_2 = \sum k(k-1) w_k, \qquad M_3 = \sum k(k-1)(k-2) w_k.$$

For $G(y) = (1-y)^{-1/2}$, equation (B.2) and its first two derivatives at $Y = Y_a$ give

$$\begin{aligned} 2aM_1 &= G(Y), \\ 2aM_2 &= YG'(Y) - (2a-1)M_1, \\ 2a(M_3 + 2M_2) &= Y^2 G''(Y) + M_2. \end{aligned} \tag{B.12}$$

In particular,

$$0 \leq \partial_u K_a(u) \leq \frac{M_2}{2} \leq \frac{YG'(Y)}{4a} \leq \frac{3\sqrt{3}}{8} < \frac{21}{32}. \tag{B.13}$$

We also need a bound for its memory derivative. Write $d_k = 2k/(2ak-1)$. Differentiating $\sum w_k = 1$ gives

$$\frac{Y_a'}{Y_a} M_1 = \sum d_k w_k \leq 6, \qquad \frac{Y_a'}{Y_a} \leq \frac{12a}{G(Y)}. \tag{B.14}$$

Here $d_k$ decreases in $k$ and $d_1 \leq 6$ on $I$. The logarithmic derivative $\partial_a \log w_k = kY_a'/Y_a - d_k$ strictly increases in $k$. Since $1 + u + \cdots + u^{k-1}$ is nondecreasing in $k$, the covariance identity for derivatives of expectations gives $\partial_a K_a(u) \geq 0$. Here the $w_k$ themselves form a probability law.

In $\partial_a \partial_u K_a$, discard the negative term $-d_k w_k$ from $\partial_a w_k$, then bound the derivative of $1 + u + \cdots + u^{k-1}$ by $k(k-1)/2$. Equations (B.12) and (B.14) give

$$\begin{aligned} \partial_a \partial_u K_a &\leq \frac{Y_a'}{Y_a} \frac{M_3 + 2M_2}{2} \\ &\leq 3\left(\frac{M_2}{G(Y)} + \frac{Y^2 G''(Y)}{G(Y)}\right) \leq 3\left(\frac{3}{4} + 3\right) = \frac{45}{4}. \end{aligned} \tag{B.15}$$

The last bounds use $M_2/G(Y) \leq Y/[4a(1-Y)]$ and $Y^2 G''(Y)/G(Y) = 3Y^2/[4(1-Y)^2]$. Since $K_a \geq 1$, $\partial_a K_a \geq 0$ and $\partial_u K_a \geq 0$, we obtain

$$\partial_a\left(\frac{\partial_u K_a}{K_a}\right) = \frac{\partial_a \partial_u K_a}{K_a} - \frac{\partial_u K_a \, \partial_a K_a}{K_a^2} \leq \frac{45}{4}.$$

Thus (B.11) satisfies the strict bound

$$\partial_u \partial_a \log W_a(u) = \frac{3}{1-u} - 3\frac{\partial_u K_a}{K_a} - s_a \partial_a \left(\frac{\partial_u K_a}{K_a}\right)$$
$$\geq 3 - \frac{63}{32} - \frac{7}{100}\frac{45}{4} = \frac{39}{160} > 0. \tag{B.16}$$

The score $\partial_a \log W_a$ is therefore strictly increasing in $u$. If $\nu_a$ denotes the normalized measure in (B.11), differentiation gives

$$\frac{d}{da}\int u\, d\nu_a(u) = \mathrm{Cov}_{\nu_a}\big(u, \partial_a \log W_a(u)\big) > 0.$$

The strict inequality follows by writing the covariance as one half the expectation of $(U-V)(S(U)-S(V))$ for independent $U, V \sim \nu_a$ and the strictly increasing score $S$. Differentiation under the integral is justified by the integrable bound $C(1-u)^{-7/100}(1+|\log(1-u)|)$ on this compact parameter interval.

It follows that

$$\mathcal{R}(a) = (1+a)^2 Y_a \int u\, d\nu_a(u)$$

has strictly positive derivative in the interior of $I$. The signs already proved and the intermediate value theorem give exactly one zero of $\mathcal{R}-1$. There are no zeros outside $I$. Finally, (B.3) writes

$$g_a''(0) = \frac{\mathcal{Z}_a}{\pi\Gamma(3-3a)}\,[\mathcal{R}(a)-1], \qquad \mathcal{Z}_a = \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a}\, dy > 0.$$

The normalizing factor is continuously differentiable near the zero, by the same rescaling and domination. At the zero its derivative contributes nothing, so $\partial_a g_a''(0) = \mathcal{Z}_a \mathcal{R}'(a)/[\pi\Gamma(3-3a)] > 0$. The shape alternatives in Theorem 3.1 now give unimodality below $a_*$ and bimodality above it. At the critical point, let $a \uparrow a_*$ from below. Joint continuity preserves $g_a(x) \geq g_a(y)$ for $0 \leq x < y$. Thus the critical density is nonincreasing on the positive half-line too. Analyticity excludes a constant interval, so its unique mode is at zero. □

## C Locating the threshold and its critical derivatives

Theorem 3.3 proves the unique transition analytically. Here we locate it and certify the derivatives used in the critical expansions. Appendix D proves the error bounds. Fix $a \in (1/2, 1)$ and write $\phi(t) = \mathbb{E}e^{it\mathcal{L}_a}$. Put $\theta_j = 1$ for even $j$ and $\theta_j = a$ for odd $j$, let $m_0 = m_1 = 1$ and

$$m_n = \frac{\sum_{j=1}^{n-1} \theta_j m_j m_{n-j}}{na - \theta_n}, \qquad c_n = \frac{m_n}{\Gamma(1+an)} \quad (n \geq 2).$$

Then $\mathbb{E}\mathcal{L}_a^n = n! c_n$ [9, Theorem 1.4], with the same definition of $c_0, c_1$. The exponential moments in [9, Corollary 2.11] give $\phi(t) = \sum_n c_n (it)^n$ for every real $t$. The fixed-point equation [9, Theorem 1.1 and (2.10)] and conjugate symmetry give

$$|\phi(t)| \leq \int_0^1 |\phi(tv^a)|\,|\phi(t(1-v)^a)|\, dv. \tag{C.1}$$

**Corollary C.1** (a certified location)**.** *The unique threshold in Theorem 3.3 satisfies* $z_- < a_* < z_+$*, where*

$$z_- = 1838418790980431/2748779069440000 = 0.668812860014597754\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.668812860014601028\ldots.$$

*The width of this interval is less than* $3.3 \cdot 10^{-15}$*.*

*Proof.* At each endpoint we integrate the real moment series and bound the remainder and Fourier tail by Lemmas D.1 to D.5. The recorded enclosures give $g''_{z_-}(0) < 0 < g''_{z_+}(0)$. Continuity and Theorem 3.3 place its unique zero between them. □

The endpoint computation bounds the series remainder by Lemma D.2 and $|\phi|$ on frequency cells by Lemma D.3. Lemma D.4 propagates $|\phi(t)| \le Ct^{-\beta}$ once its explicit test $R < 1$ succeeds. Lemma D.5 bounds the omitted curvature integral. The two endpoint checks use real-parameter moments directly, independently of the parameter Taylor models. Their records and exact inputs are in Appendix G.

To enclose derivatives over a memory interval, we expand in $a - a_0$ before taking absolute values. This preserves cancellation between large moment terms. Proposition D.7 bounds the Taylor remainder using a complex moment majorant. Section D.3 differentiates the smoothing equation and proves the uniform derivative-tail bounds. On $[0.66881285, 0.66881287]$ the resulting enclosures give

$$0.68 < \partial_a g''_a(0) < 0.70, \qquad -0.09 < g^{(4)}_a(0) < -0.07.$$

The refined evaluation in Table 1 uses a smaller neighborhood containing $[z_-, z_+]$. These numerical bounds determine the quartic top and its constants. They are not inputs to the analytic uniqueness proof.

## D Bounds for the numerical enclosures

We prove the bounds used in Appendix C, using its notation and the smoothing inequality (C.1). These locate the threshold of Theorem 3.3 and certify the critical derivatives. Write $\Phi_N(t) = \sum_{n \le N} c_n (it)^n$ for a partial sum of the moment series.

### D.1 Bounds at a fixed memory

**Lemma D.1** (moment bounds)**.** *Let $N \ge 3$.*

*(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.*

*(b) Let $S \subset \mathbb{C}$ be compact, with $\frac{1}{2} < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with $z$ in place of $a$, and $c_n(z) = m_n(z)/\Gamma(1+nz)$. Then every $m_n$ is analytic near $S$. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on $S$ for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$*

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde{A}\tilde{B}^n}{\Gamma(1+n\alpha)}, \qquad \tilde{A} = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde{B} = Be^{\rho^2/(2\alpha)}.$$

*Proof.* For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in $n$, since its derivative is $(1-a)/(n-1)^2$, and it equals $A$ at $n = N+1$. So $m_n \le AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \ge n\alpha - 1 > 0$ on $S$, because $\alpha \le 1$. So $m_n$ is analytic near $S$, and the induction of (a) goes through with $|\theta_j| \le 1$. For $x > 0$ and real $y$,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k \ge 0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \le \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [19, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \ge 1 + n\alpha$ and $|y| \le n\rho$. Then $y^2/x^2 \le \rho^2/\alpha^2$ and $y^2/x \le n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \ge \Gamma(1+n\alpha)$, as $\Gamma$ increases on $[1.47, \infty)$. □

**Lemma D.2** (truncation)**.** *Let $0 < \alpha \le 1$, and suppose that $|c_n| \le \tilde{A}\tilde{B}^n/\Gamma(1+n\alpha)$ for $n > N$. Let $t > 0$, $0 \le k \le N$ and $n_0 = N+1$, and put*

$$u = \frac{\tilde{A}\tilde{B}^{n_0} n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)}, \qquad r = \tilde{B}t\Big(1+\frac{1}{n_0}\Big)^k \frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

*If $r < 1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N} c_n (it)^n\big| \le \tau_k(t) := u/(1-r)$, and $\tau_k$ increases on $(0,t]$.*

*Proof.* The $k$th derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde{A}\tilde{B}^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde{B}t(1+1/n)^k\Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in $n$, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term $u$ and ratio $r$. Both $u$ and $r$ increase in $t$. □

**Lemma D.3** (frequency cells)**.** *Let $\nu$ be even, $t_c > 0$ and $|\delta| \le h$. Then*

$$|\phi(t_c+\delta)| \le \sum_{k<\nu} |\phi^{(k)}(t_c)| h^k/k! + c_\nu h^\nu.$$

*Proof.* This is Taylor's formula, with $|\phi^{(\nu)}| \le \mathbb{E}|\mathcal{L}_a|^\nu = \mathbb{E}\mathcal{L}_a^\nu = \nu!\, c_\nu$. □

**Lemma D.4** (Fourier tail)**.** *Let $0 < T_2 < T_3$ with $T_3 \ge 2^a T_2$, and let $\beta > 3$ with $a\beta > 1$. Cover $[0,T_3]$ by cells $[l_j, r_j]$ with $l_0 = 0$ and with $T_2$ among the end points, and let $s_j \ge \sup_{[l_j,r_j]}|\phi|$. Let $C$ be the largest $s_j r_j^\beta$ over the cells that meet $[T_2,T_3]$, and put*

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

*If $R < 1$, then $|\phi(t)| \le Ct^{-\beta}$ for every $t \ge T_3$.*

*Proof.* Let $b = s_j$ on the cells in $[0,T_3)$ and $b(t) = Ct^{-\beta}$ for $t \ge T_3$. On a cell that meets $[T_2,T_3]$ we have $s_j \le C r_j^{-\beta} \le Ct^{-\beta}$. So $b(t) \le Ct^{-\beta}$ for $t \ge T_2$, and $|\phi| \le b$ on $[0,T_3]$. Let $J(t) = \int_0^1 b(tv^a)\, b(t(1-v)^a)\, dv$.

We show that $J(t) \le Rb(t)$ for $t \ge T_3$. By symmetry $J(t)$ is twice the integral over $v \le \frac{1}{2}$. There $t(1-v)^a \ge 2^{-a}T_3 \ge T_2$, so $b(t(1-v)^a) \le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\, dv$. Put $v_0 = (T_2/t)^{1/a} \le \frac{1}{2}$. On $v < v_0$ we substitute $u = tv^a$, and this part becomes

$$\frac{2}{a} t^{-1/a} \int_0^{T_2} s(u)\big(1-(u/t)^{1/a}\big)^{-a\beta} u^{1/a-1}\, du.$$

Here $t^{-1/a} \le \frac{1}{2}T_2^{-1/a}$ and $(u/t)^{1/a} \le \frac{1}{2}(u/T_2)^{1/a}$. The weight increases in $u$, so on each cell we take it at $r_j$. As $\int_l^r u^{1/a-1}du = a(r^{1/a} - l^{1/a})$, this part is at most the first term of $R$. On $v \ge v_0$ we use $b(tv^a) \le Ct^{-\beta}v^{-a\beta}$ and $(1-v)^{-a\beta} \le 2^{a\beta}$. This part is at most $2^{1+a\beta}T_2^{-\beta}(T_2/t)^{1/a}C/(a\beta-1)$, which is at most the second term of $R$.

Let $\varsigma$ be the supremum of the $\tau \ge T_3$ with $|\phi| \le b$ on $[0,\tau]$, and suppose that $\varsigma < \infty$. By continuity $|\phi| \le b$ on $[0,\varsigma]$. The arguments in (C.1) at $t = \varsigma$ lie in $[0,\varsigma]$, so $|\phi(\varsigma)| \le J(\varsigma) \le Rb(\varsigma) < b(\varsigma)$. As $|\phi|$ and $b$ are continuous on $[T_3,\infty)$, we get $|\phi| < b$ near $\varsigma$, against the choice of $\varsigma$. □

This is an explicit form of Step 4 in the proof of [9, Theorem 2.5], with computed constants.

**Lemma D.5** (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then $g_a$ has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big( \sum_{\substack{n\le N\\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E\Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta-3}.$$

*Proof.* As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi}\int e^{-itx} \operatorname{Re}\phi(t)\,dt$, and dominated convergence allows two derivatives under the integral [18, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\,dt$. On $[0, T_3]$ we replace $\phi$ by $\Phi_N$, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma D.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\,dt \le \frac{1}{3}T_3^3\tau_0(T_3)$. The integral over $t > T_3$ is at most $CT_3^{3-\beta}/(\beta-3)$. □

### D.2 Bounds on memory intervals

Near the threshold, the series for $\phi$ cancels terms of size about $10^{60}$. Direct interval evaluation would need intervals in $a$ narrower than about $10^{-65}$. We instead expand in $a$ about one exact point, so the cancellation occurs before we bound the remainder.

**Lemma D.6** (Cauchy remainder)**.** *Let $h$ be analytic near the closed disc $|z - a_0| \le \rho$, with $|h| \le M$ there and Taylor coefficients $h_i$ at $a_0$. If $0 < \vartheta < 1$ and $|a - a_0| \le \vartheta\rho$, then $|h(a) - \sum_{i\le K} h_i (a-a_0)^i| \le M\vartheta^{K+1}/(1-\vartheta)$.*

*Proof.* Cauchy's estimate gives $|h_i| \le M\rho^{-i}$. Sum the geometric tail. □

Let $I = [a_0 - w, a_0 + w]$, $0 < \vartheta < 1$ and $\rho = w/\vartheta$, and let $S$ be the square $a_0 + [-\rho, \rho] + i[-\rho, \rho]$, which contains the disc of Lemma D.6. Suppose that $\alpha = a_0 - \rho > \frac{1}{2}$, that $|z| \le 1$ on $S$, and that $\tilde{A}, \tilde{B}$ are as in Lemma D.1(b). Let $\tau_k$ be the bound of Lemma D.2 with these constants. It holds for every real $a \in I$. For a cell with center $t_c$ write $\partial_t^k \Phi_N(z, t_c) = E_k(z) + iO_k(z)$, where $E_k$ is the sum over even $n$ and $iO_k$ the sum over odd $n$. Both $E_k$ and $O_k$ are real for real $z$. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at $a_0$, and put $M_k = \sum_{n\le N} \sup_S |c_n|\, n(n-1)\cdots(n-k+1)\, t_c^{n-k}$.

**Proposition D.7** (enclosure on an interval)**.** *(a) For every $a \in I$ and every $t$ in the cell with center $t_c$ and half-width $h$,*

$$|\phi_a(t)| \le \sum_{k<12} \Big( \sum_{i\le K} w^i \sqrt{E_{k,i}^2 + O_{k,i}^2} + \frac{M_k \vartheta^{K+1}}{1-\vartheta} + \tau_k(t_c)\Big)\frac{h^k}{k!} + h^{12} \sup_S |c_{12}|.$$

*(b) Let $T_2$, $T_3 \ge 2^{a_0+w} T_2$, $\beta$ and the cells be as in Lemma D.4, with $s_j$ the bound of (a), and suppose that $R < 1$ for every $a \in I$. Put*

$$F(z) = \sum_{n\le N,\, n \text{ even}} (-1)^{n/2} c_n(z) T_3^{n+3}/(n+3).$$

*Let $F_i$ be its Taylor coefficients at $a_0$, and let $M_F$ be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a \in I$*

$$\Big| g_a''(0) + \frac{1}{\pi}\sum_{i\le K} F_i (a-a_0)^i \Big| \le \eta = \frac{1}{\pi}\Big( \frac{M_F \vartheta^{K+1}}{1-\vartheta} + \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta-3}\Big),$$

*and $a \mapsto g_a''(0)$ is continuous on $I$.*

*Proof.* For (a) we apply Lemma D.3 with $\nu = 12$, and Lemma D.2 to replace $\phi_a^{(k)}$ by $\partial_t^k \Phi_N$. For real $a$ we have $|\partial_t^k \Phi_N(a, t_c)| = |(E_k(a), O_k(a))|$. Lemma D.6 for $E_k$ and $O_k$,

with $\sup|E_k| + \sup|O_k| \le M_k$ on $S$, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a) \le \sup_S |c_{12}|$.

For (b), $T_3 \ge 2^a T_2$ for every $a \in I$, so Lemma D.4 applies at each $a$. Then Lemma D.5 and Lemma D.6 for $F$ give the enclosure. By Lemma D.1(b) the series for $\phi_a(t)$ converges uniformly for $a \in I$ and $t$ in compact sets. So $\phi_a(t)$ is continuous in $a$. Moreover $t^2|\phi_a(t)|$ is at most $t^2$ on $[0, T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a \in I$. Dominated convergence gives the continuity. □

### D.3 Critical derivative bounds

To compute the memory derivative and critical constants, we differentiate the smoothing equation and then bound the tail. The error bound in Lemma D.5 alone gives no bound on the derivative of that error.

**Lemma D.8** (differentiated smoothing bound)**.** *Fix a compact memory interval $I \subset (1/2, 1)$. Suppose that the cell certificate of Lemma D.4 gives, uniformly on $I$, $f(t) := |\phi_a(t)| \le Ct^{-\beta}$ for $t \ge T_2$, and*

$$H_\beta(t) := 2\int_0^{1/2} f(tv^a)(1-v)^{-a\beta}\,dv \le R < 1 \quad (t \ge T), \qquad T \ge 2^a T_2.$$

*Let $\beta > 4$, $\gamma = \beta - 1$, $a\gamma > 1$, and $u = |\partial_t \phi_a|$, $w = |\partial_a \phi_a|$. Suppose that $u \le M$ everywhere, $w \le M_a$ on $[0, T_2]$, and cells give $u(t) < D_1 t^{-\beta}$ and $w(t) < D_a t^{-\gamma}$ on $[T_2, T]$. Put*

$$A_1 = \frac{MC2^{a(\beta-1)}}{a+1}, \qquad J = 2C2^{a\beta}T^{\gamma-\beta-1/a},$$

$$\kappa = R + \frac{JT_2^{1/a-\gamma}}{a\gamma - 1}, \qquad A_0 = JM_aT_2^{1/a},$$

$$A_S = MC2^{a(\beta-1)}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\} + D_1(\log 2)R + CR/T.$$

*If $A_1 + RD_1 < D_1$ and $A_0 + A_S + \kappa D_a < D_a$, with $\kappa < 1$, then the two derivative bounds hold for every $t \ge T_2$. Consequently, in the interior of $I$,*

$$\frac{d}{da}g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt, \qquad \left|\frac{1}{\pi}\int_T^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt\right| \le \frac{D_aT^{3-\gamma}}{\pi(\gamma-3)}. \tag{D.1}$$

*Proof.* The complex majorant in Lemma D.1(b) makes the moment series locally uniformly convergent in the frequency and memory. Cauchy's estimate on a smaller parameter disc gives the same conclusion for its memory derivative and every fixed frequency derivative. We may therefore differentiate the series for $\phi_a$ term by term. The smoothing equation, with $p = (1+a)/2$, is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a)\{p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a)\}\,dv.$$

On compact frequency sets, its differentiation under the integral is justified by local boundedness of these derivatives and boundedness of $v^{\inf I}|\log v|$. Conjugate symmetry gives even moduli. Writing $x = tv^a$ and $y = t(1-v)^a$, we obtain

$$u(t) \le \int_0^1 \{v^a u(x)f(y) + (1-v)^a f(x)u(y)\}\,dv,$$

$$w(t) \le \int_0^1 \{w(x)f(y) + f(x)w(y)\}\,dv + S(t),$$

$$S(t) \le \int_0^1 \{x|\log v|u(x)f(y) + y|\log(1-v)|f(x)u(y) + f(x)f(y)\}\,dv.$$

The final term includes the derivative of $p$. Its coefficient is one because $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \le 2f(y)$.

Split each integral at $1/2$ and reflect its second half. For the frequency derivative, the smaller-argument term is at most $A_1 t^{-\beta}$, by $u \le M$ and

$$2\int_0^{1/2} v^a (1-v)^{-a\beta}\, dv \le \frac{2^{a(\beta-1)}}{a+1}.$$

The larger-argument term is at most $RD_1 t^{-\beta}$. For the memory derivative, split the smaller-argument term again at $v_0 = (T_2/t)^{1/a}$. Its bound is

$$2C2^{a\beta} t^{-\beta-1/a} \left\{ M_a T_2^{1/a} + \frac{D_a T_2^{1/a-\gamma}}{a\gamma - 1} \right\}.$$

The larger-argument term is at most $RD_a t^{-\gamma}$, since $H_\gamma \le H_\beta$. As $\gamma - \beta - 1/a < 0$ and $t \ge T$, these two terms are bounded by $(A_0 + \kappa D_a) t^{-\gamma}$. In $S$, the smaller logarithmic term uses

$$\int_0^{1/2} v^a (-\log v)\, dv = 2^{-a-1} \left\{ \frac{\log 2}{a+1} + \frac{1}{(a+1)^2} \right\};$$

the larger uses $-\log(1-v) \le \log 2$ and $H_\beta \le R$. The mixing-weight term is at most $CRt^{-\beta}$. Thus $S(t) \le A_S t^{-\gamma}$.

To extend the bounds beyond the cells, maximize the continuous weighted derivative on any $[T_2, U]$. If its maximum lies in $[T_2, T]$, use the cell bound. Otherwise repeat the preceding inequalities with that maximum in place of the proposed constant, and rearrange the strict contraction inequality. This first proves the frequency bound, then the memory bound, independently of $U$. The integrable uniform bound $t^2 w(t) \le D_a t^{2-\gamma}$ now permits differentiation of the Fourier formula for $g_a''(0)$ on the interior of $I$ and gives (D.1). □

The computation uses

$$I = [0.66881285, 0.66881287], \quad T_2 = 6, \quad T = 10, \quad \beta = 8, \quad \gamma = 7,$$

and the conservative constants

$$C = 17/1000, \quad R = 83/1000, \quad M = 2, \quad D_1 = 2, \quad M_a = D_a = 4.$$

It checks $A_1 < 1$, $\kappa < 1/10$ and $A_0 + A_S < 2$, so both contractions have strict slack. The resulting tail bound is $1/(10000\pi) < 1/30000$. The certified zero bracket lies strictly inside $I$.

For the finite-frequency integral, let $w_0$ be the halfwidth of $I$, $\rho > w_0$ the Cauchy radius and $\vartheta = w_0/\rho$. Differentiating the finite Taylor polynomial in memory leaves the coefficient remainder factor

$$\frac{1}{\rho} \sum_{i>K} i\vartheta^{i-1} = \frac{\vartheta^K \{(K+1) - K\vartheta\}}{\rho(1-\vartheta)^2}.$$

The omitted moment series has derivative bound $\tau_k(t)/(\rho - w_0)$ by Cauchy's estimate. On each frequency cell we use a Taylor expansion of order $48$, with its remainder bounded by the complex coefficient majorant. (At complex memory we cannot use a probability moment.) The signed interval checks give

$$0.68 < \frac{d}{da} g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I). \tag{D.2}$$

The second enclosure uses the same Fourier bound with weight $t^4$, whose tail is $CT^{-3}/(3\pi)$. The file `certify_transition.py` performs these checks, including the initial derivative-cell bounds. The recorded run uses $1100$ moments and $1600$-bit precision under `python -O`. Each check uses `require`, so it also runs with optimization enabled.

At the unique zero from Theorem 3.3, evenness and the second inequality in (D.2) give

$$g'_{a_*}(x) = \tfrac{1}{6} g^{(4)}_{a_*}(0)x^3 + o(x^3) < 0 \quad \text{for all sufficiently small } x > 0.$$

The fourth derivative is nonzero, so the critical top is quartic. The analytic theorem already gives the unimodal critical shape and excludes other curvature zeros on $(1/2, 1)$.

**The critical constants.** For Table 1 we apply the same estimates on $I_* = [z_- - 10^{-14}, z_- + 10^{-14}]$, where $z_- = 1838418790980431/2748779069440000$ is the lower endpoint of the certified threshold bracket. The entire bracket lies inside $I_*$. Take

$$T_2 = 10, \quad T = 16, \quad \beta = 12, \quad \gamma = 11,$$
$$C = 1/1000, \quad R = 1/20, \quad M = D_1 = 2, \quad M_a = D_a = 4.$$

The base Fourier bound, initial derivative cells and both strict contraction inequalities hold uniformly on $I_*$. We use $2000$ moments, $2600$-bit ball arithmetic, Cauchy ratio $1/10000$, frequency halfwidth $1/32$ and frequency Taylor order $48$. The omitted tails for $C_*$, $B_*$ and $H_*$ are bounded respectively by

$$\frac{D_a T^{3-\gamma}}{\pi(\gamma - 3)}, \qquad \frac{CT^{5-\beta}}{\pi(\beta - 5)}, \qquad \frac{CT^{2-\beta}}{\pi(\beta - 2)}.$$

The finite-frequency integrals use the same parameter and moment remainders as above, with weights $t^2$, $t^4$ and $t$. The bounds hold throughout the interval, so they apply at $a_*$ without needing its exact value. Outward rounding then gives the two derived coefficients in the table. The record and rerun command are in Appendix G.

## E Signs of the second difference

We prove the sign bounds of Section 2.

### E.1 Averaging adjacent entries

Insert each $\eta_i$ between its two neighbors in $\zeta$. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so $\eta_i$ is placed next to an entry of its own sign, and the pattern does not change. The same argument works when $\eta_i < 0$, proving the first claim. For the second, obtain $\eta$ from $w$ by deleting the entries $\zeta_i$. Deletion can only remove signs from the pattern. □

### E.2 Signs at the first lattice point

Put $\xi_t = P_t(1)/P_t(0)$. By (2.1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1 - \varepsilon - \frac{a}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{a\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/a$ for $t \geq 2$. For $t = 2$ this holds because $a < 1 - \varepsilon$. If it holds for $t$, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{a}$, and it holds for $t+1$. So $\xi_t$ is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq a/\varepsilon$.

Now (2.2) with $k = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{a}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - a = \varepsilon > 0$. If $t < a/\varepsilon$, both factors of $\sigma_t$ are negative and $\sigma_t > 0$. If $t > a/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because $\xi_t$ increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of $\Pi_3$. If $\varepsilon \geq \frac{1}{3}$, then $a/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-, +)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (2.1) and (2.2) use only that step $t + 1$ is $+1$ with probability $r_t(k)$ given $A_t = k$, so both hold for this $\varepsilon$. We compute $P_3(0) = \frac{1}{2}(1 - \varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1 - \varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2.2) with $t = 3$ and $k = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{a}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With $P_t$ defined by (2.1), the same computation works for every real $\varepsilon$, because the other root $1 + 2^{-1/2}$ of $s_3$ is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

### E.3 At most five sign blocks

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

To apply Lemma 2.2, we check how $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ for $3 \leq t \leq N - 1$. The first $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N - t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t - 2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t - 1$. By (2.2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big)D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - a = \varepsilon$, since $2 \leq b \leq t - 1$. The last $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 2.2, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 2.4 the sign pattern $p$ of $(s_3, \dots, s_N)$ is a subsequence of $\Pi_3$, so it is one of the lists $()$, $(+)$, $(-)$, $(+, -)$, $(-, +)$ and $(+, -, +)$. The sign pattern of $\mathcal{Z}_3$ is $p$ reversed, followed by $p$, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-, +, -)$, $(+, -, +)$ and $(+, -, +, -, +)$, and each is a subsequence of $\Pi_5$.

□

## F Derivative limits for a fixed start

The strict slope at zero needs convergence of first differences. We prove it using the higher-difference recurrence of [12, (4.10)]. The argument works at every fixed order and for any prescribed finite initial history.

**Lemma F.1.** *Fix $a \in (1/2, 1)$ and a finite initial history. Suppose the rescaled count law converges weakly to a law with smooth density $f$. For each fixed integer $r \geq 0$, linearly interpolate*

$$\frac{n^{(r+1)a}}{2^{r+1}} \nabla^r P_n(k) \quad \text{at} \quad (2k - n - r)n^{-a}, \qquad \nabla P_n(k) = P_n(k) - P_n(k-1).$$

*These interpolants converge locally uniformly to $f^{(r)}$.*

*Proof.* Repeated subtraction in (2.1) gives

$$\nabla^r P_{n+1}(k) = \left(\frac{1+a}{2} - \frac{ak}{n}\right) \nabla^r P_n(k) + \left(\frac{1-a}{2} + \frac{a(k-r-1)}{n}\right) \nabla^r P_n(k-1).$$

For sufficiently large $n$, all coefficients multiplying nonzero entries are nonnegative. Interior row sums are $1 - (r+1)a/n$. At either outer endpoint the only nonzero term has coefficient $(1+a)/2$, which is no larger once $n$ is large. Iteration gives

$$\sup_k |\nabla^r P_n(k)| \leq C_{a,r} n^{-(r+1)a}.$$

The bounds at orders $r$ and $r+1$ give uniform bounds and Lipschitz constants for the interpolants. Every subsequence therefore has a further subsequence converging locally uniformly. To identify its limit, sum by parts against a smooth compactly supported function. Weak convergence then gives the distributional derivative of order $r$ of the limiting law. The lattice spacing is $2n^{-a}$, so the displayed normalization is exactly its power $-(r+1)$. Since $f$ is smooth, this distributional derivative is $f^{(r)}$. Every subsequence has the same limit, proving the assertion. □

## G Data and certificate records

The tagged repository and submission archive contain the certificate code, records, source hashes and rerun commands. Threshold uniqueness uses no numerical record. The endpoint signs are in data/critical_shape_check.json. The critical derivative records use 1100 moments at 1600 bits, Python 3.12.14 and python-flint 0.8.0.

With python-flint 0.9.0, check_critical_shape.py checks the real threshold endpoints using 1800 moments at 2000 bits and the conditional slope using 1200 moments. cert ify_critical_constants.py encloses the critical constants using 2000 moments at 2600 bits. Both checks retain their acceptance tests under `python -O`. The numerical phase illustration and finite-length examples are not certificates. The README lists each script and its output.

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